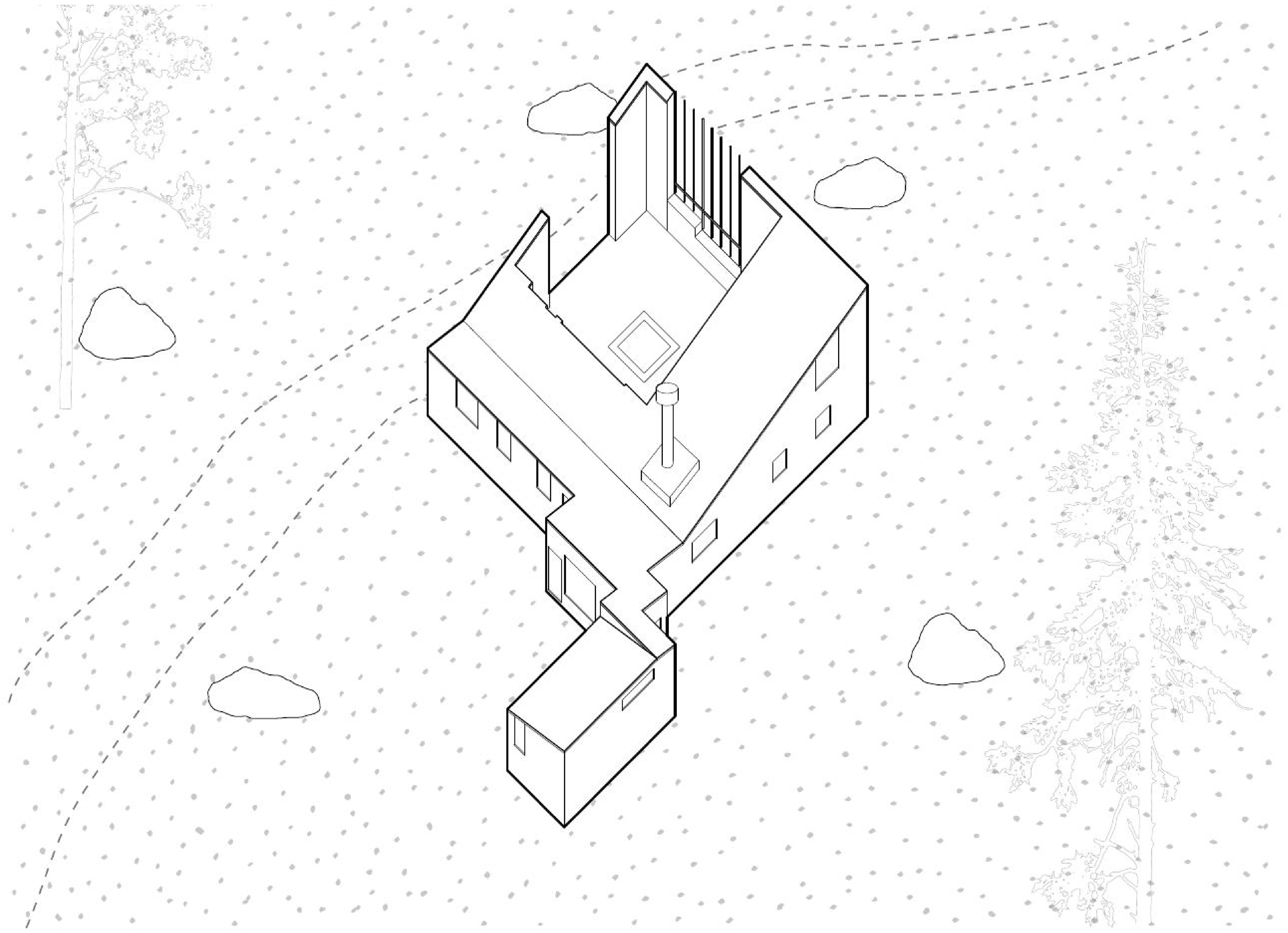
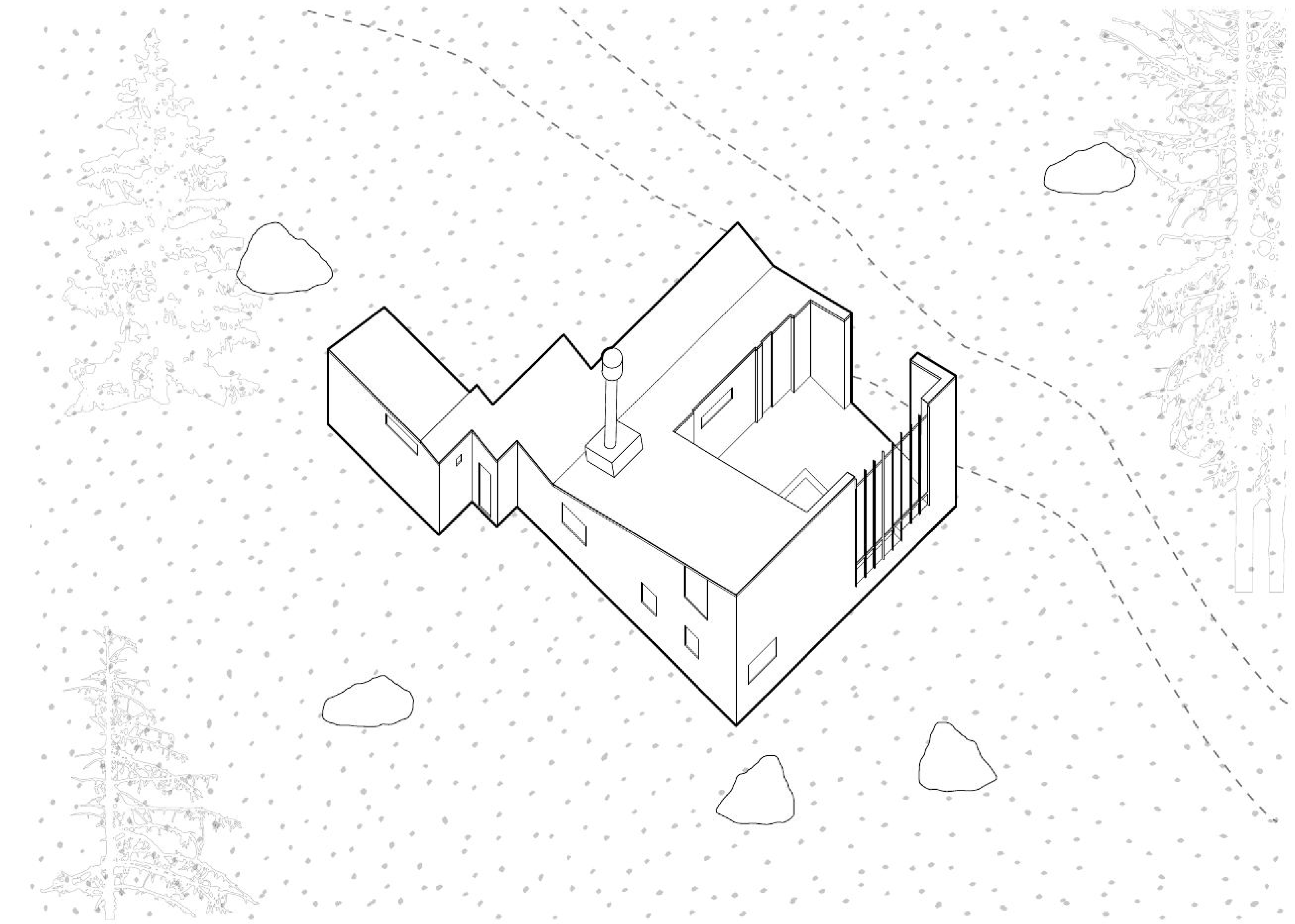


AXONOMETRIC NE - 1:200



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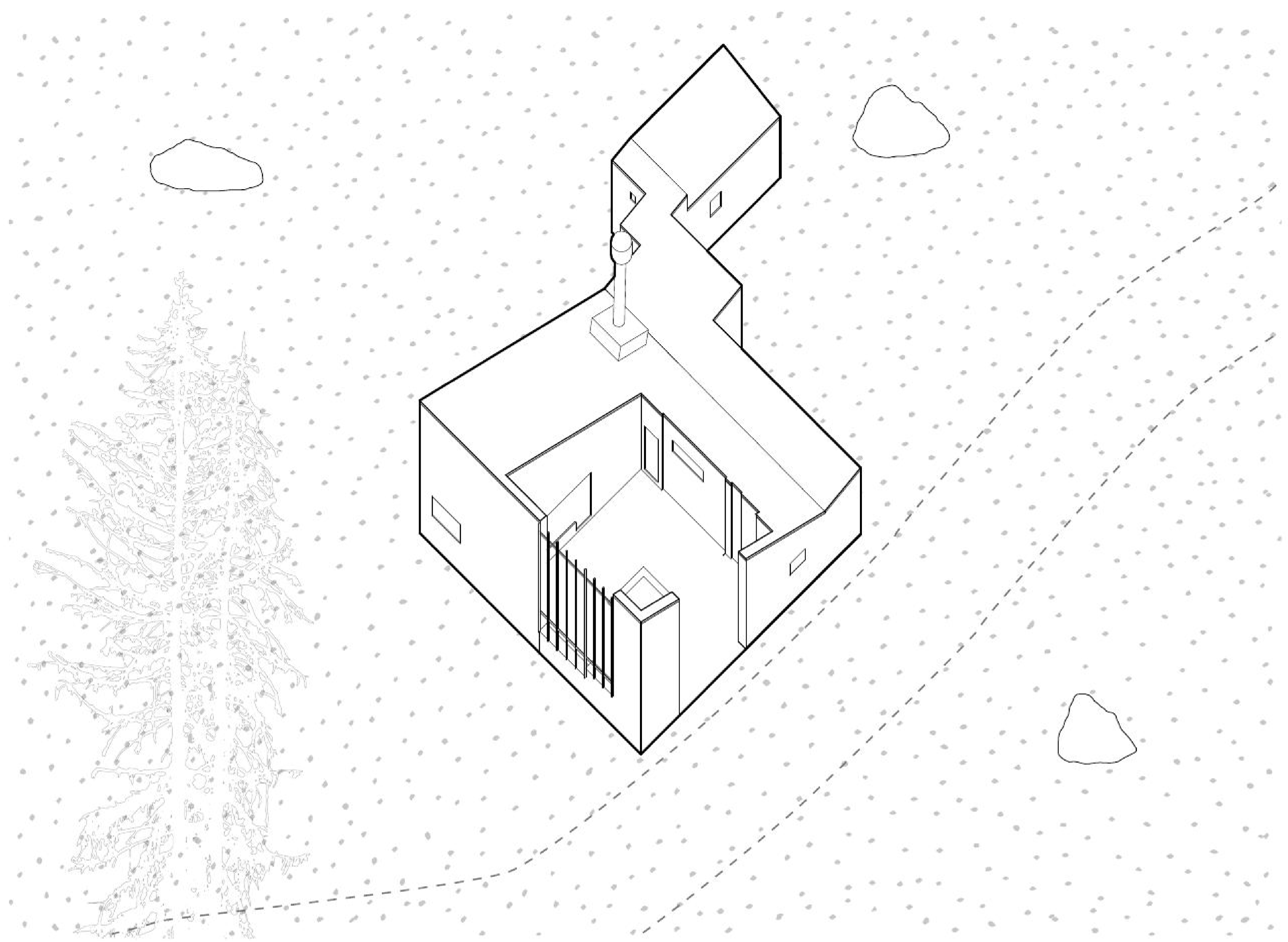

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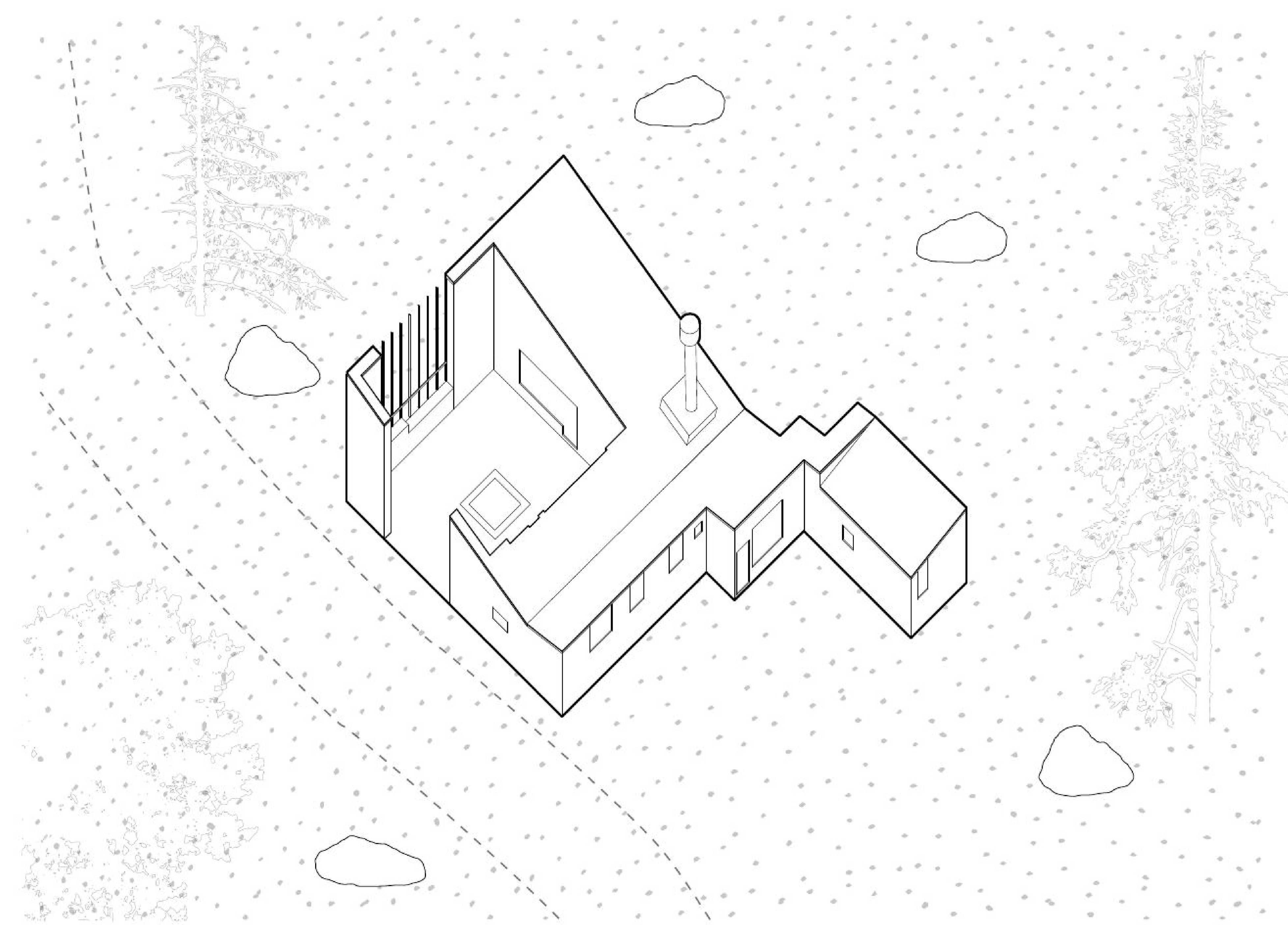
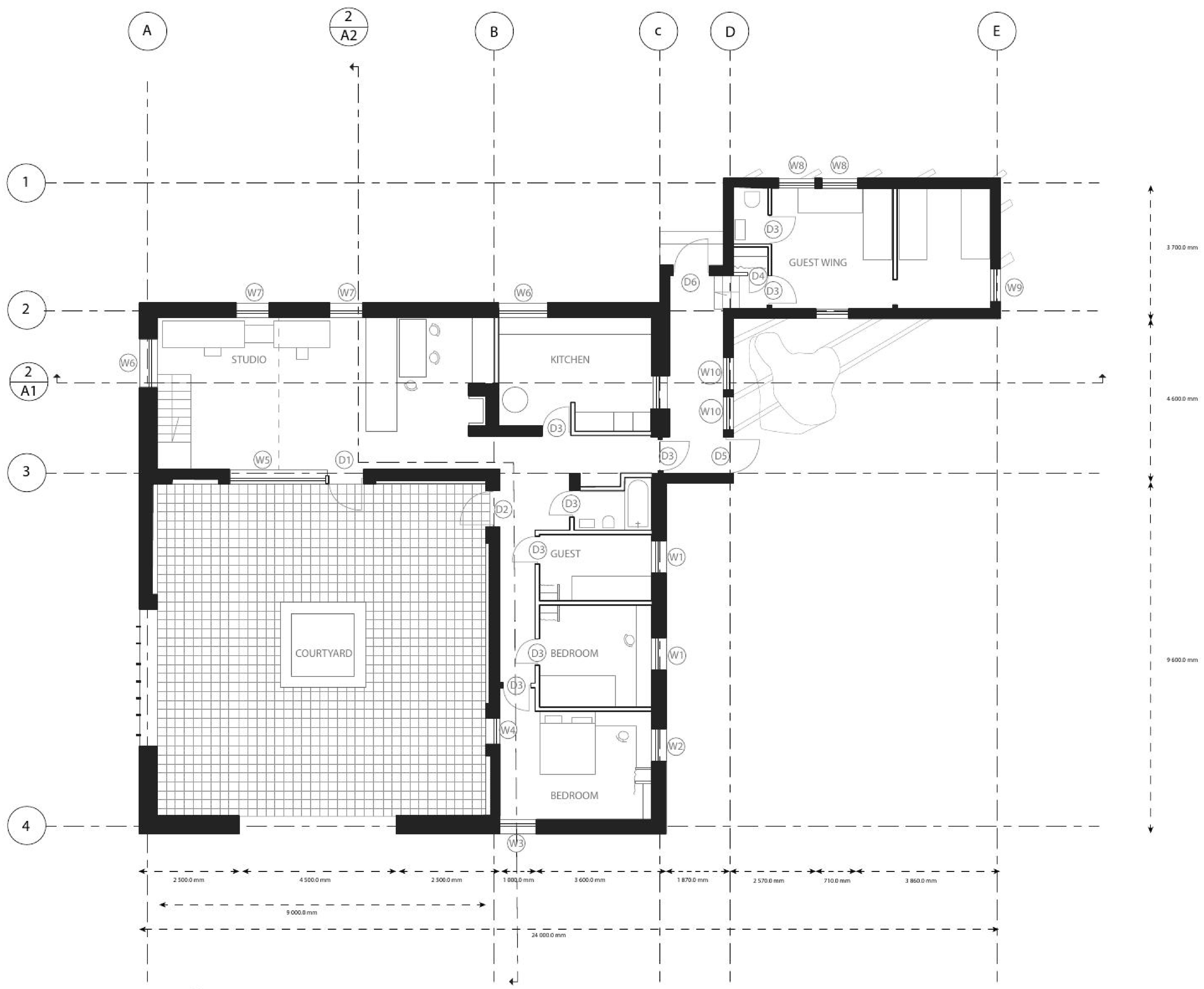
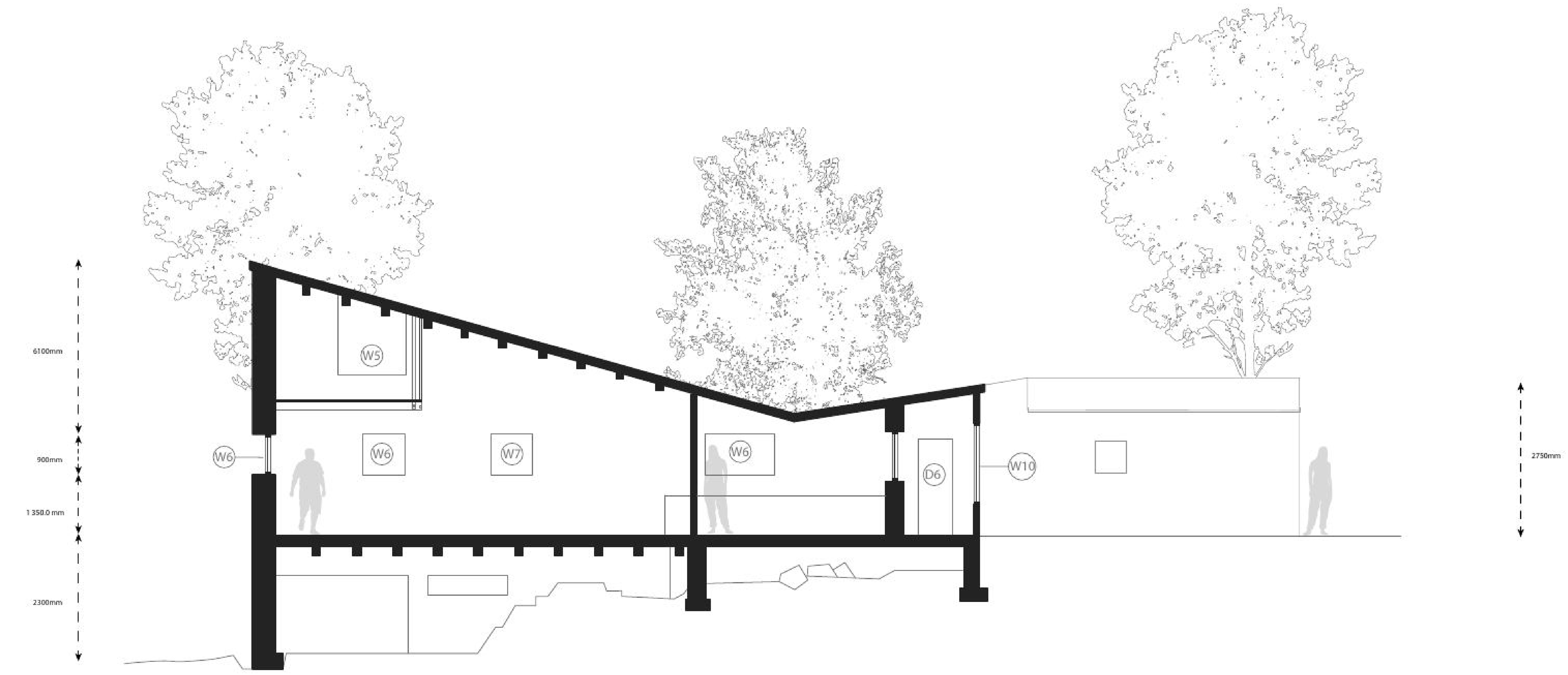


AXONOMETRIC SE - 1:200

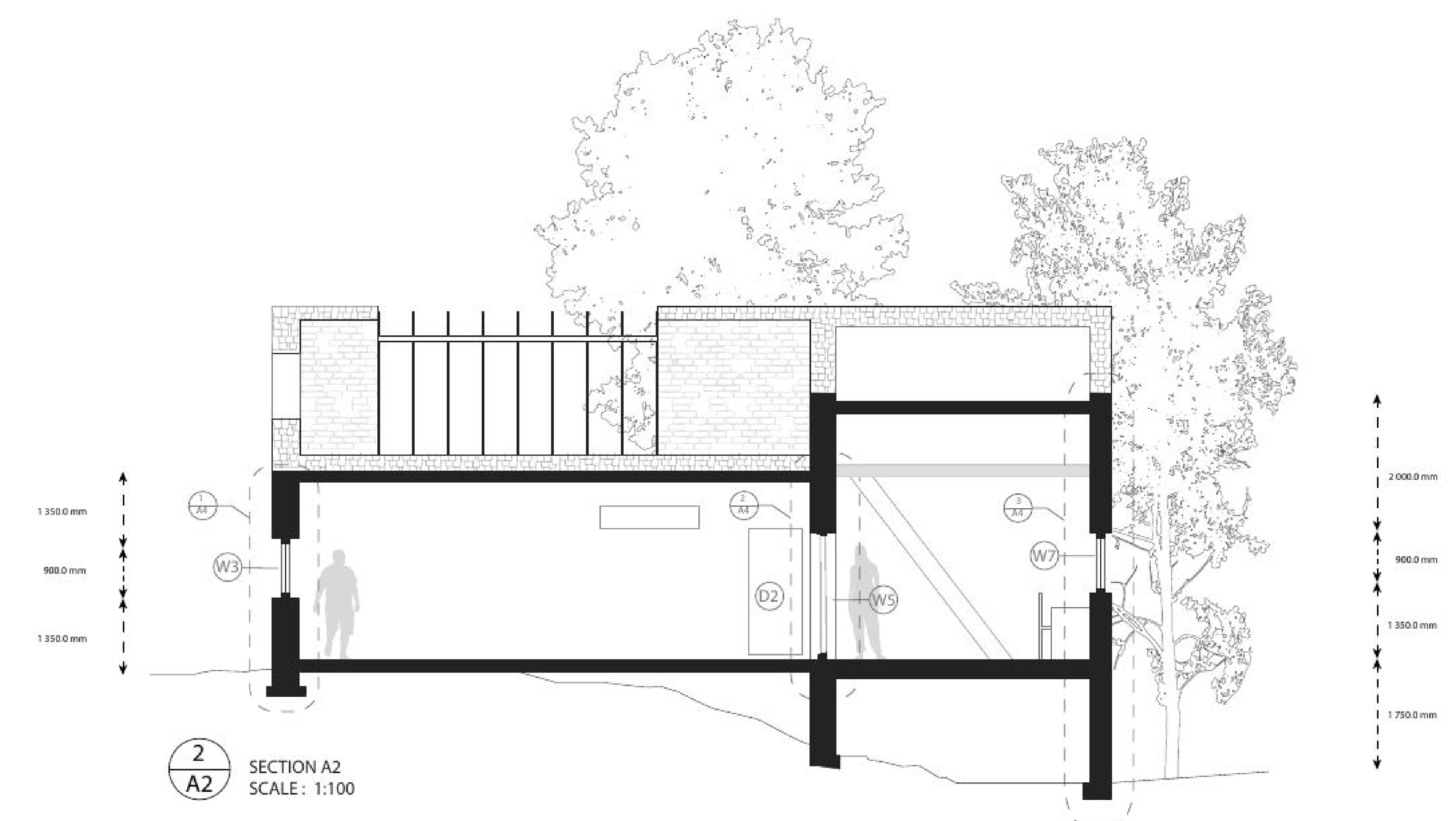
Carleton University Azrieli School of Architecture and Urbanism	EXPERIMENTAL HOUSE MELALAMMETIE,40900, FINLAND DATE OF CONSTRUCTION 1952-21954	CLIMATE AND ENVIRONMENTAL RETROFIT GROUP 7	SHEET NUMBER: A - 101 BUILDING AXONOMETRIC	TEAM MEMBER: OWEN KEEPING	REVISION SCHEDULE: Owen Keeping: 9/23/2025 - (Created digital massing model) Owen Keeping: 9/23/2025 - (Created line drawings from digital model) Owen Keeping: 10/12/2025 - (adjusted line weights in accordance to TA feedback) Owen Keeping: 10/13/2025 - (Finalized compositions, Added details) Owen Keeping: 10/14/2025 - (Revised for consistency in Graphics across drawings sheets.)



PLAN
SCALE : 1:100

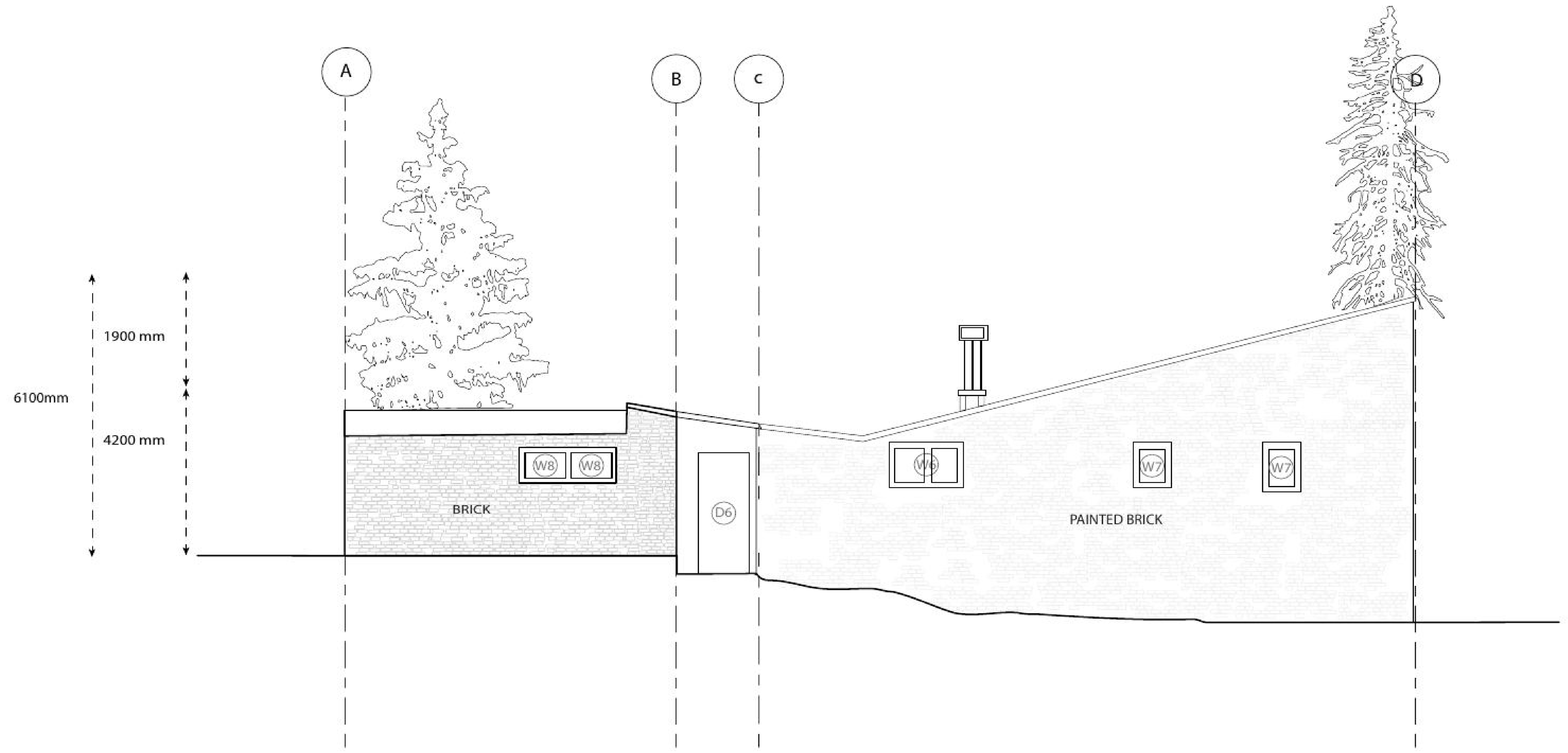


SECTION A1
SCALE : 1:100

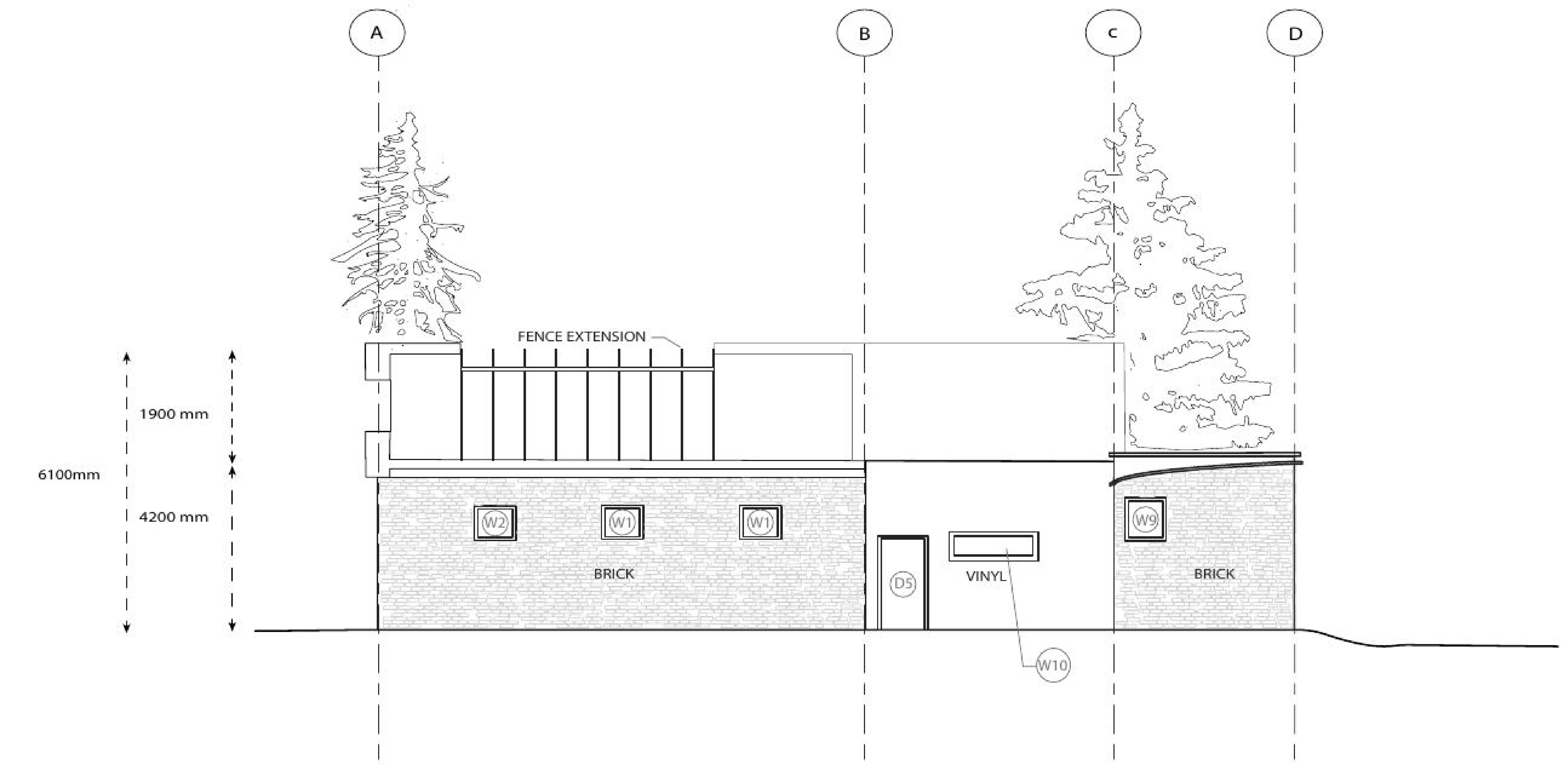


SECTION A2
SCALE : 1:100

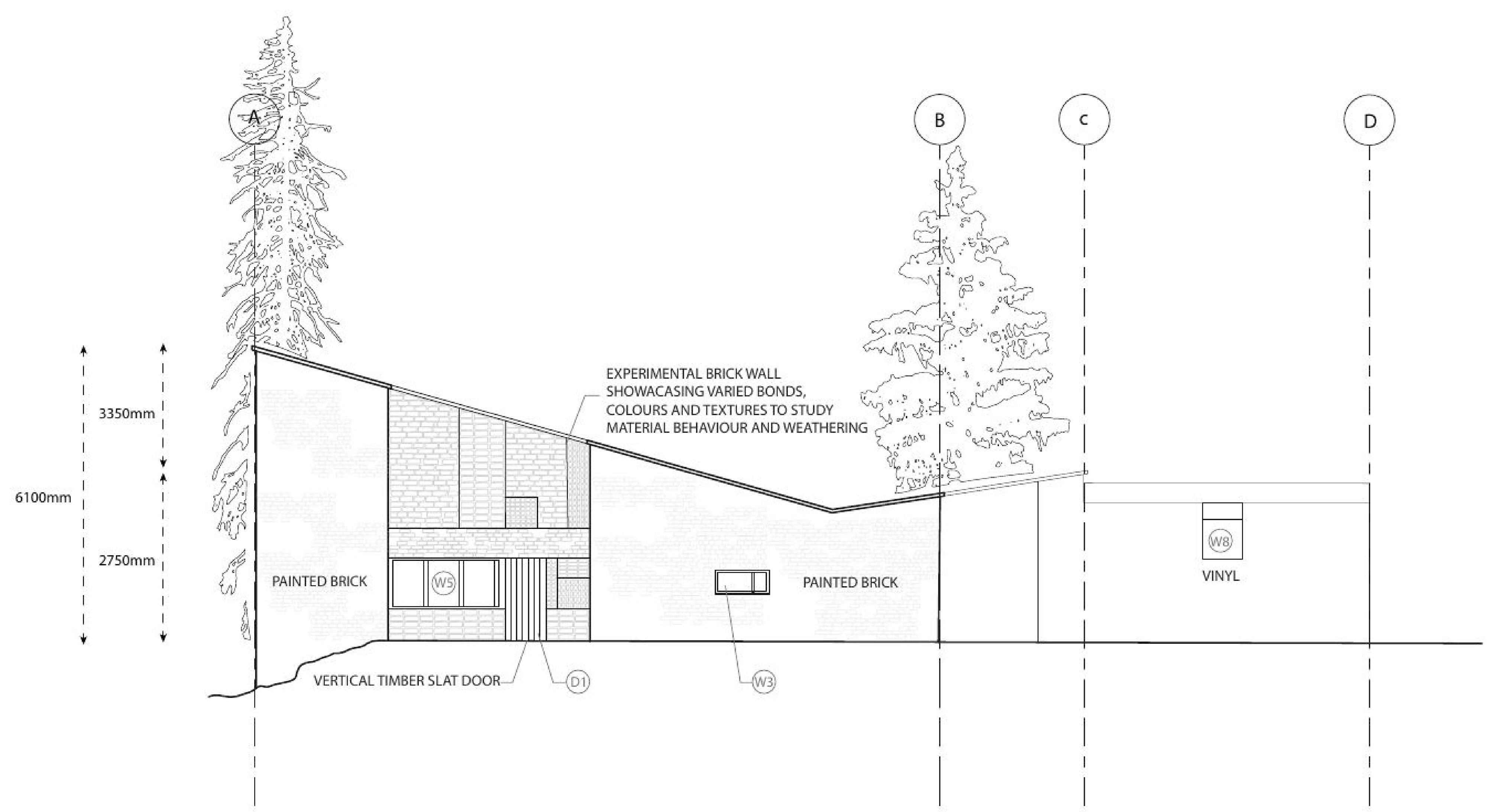
Carleton University Azrieli School of Architecture and Urbanism	EXPERIMENTAL HOUSE MELALAMMETIE,40900, FINLAND DATE OF CONSTRUCTION 1952-21954	CLIMATE AND ENVIRONMENTAL RETROFIT GROUP 7	SHEET NUMBER: A - 201 PLANS & SECTIONS	TEAM MEMBER: OWEN KEEPING	REVISION SCHEDULE: Colin Murray: 9/17/2025 - 10/14/2025 (Created Revit model) Colin Murray: 10/01/2025 - (Modified Revit model in accordance to TA feedback) Colin Murray: 9/17/2025 - 9/24/2025 (Created preliminary plan drawing) Owen Keeping: 9/24/2025 - (Used model cuts & preliminary drawings provided by a team member to enhance plan and section drawings.) Owen Keeping: 10/1/2025 - (Refined linework, adjusted graphics, and added detailing.) Owen Keeping: 10/7/2025 - (added annotations, dimensions, and tags for clarity.) Owen Keeping: 10/13/2025 - (revised annotations, dimensions, and tags.) Owen Keeping: 10/14/2025 - (Revised for consistency in Graphics across drawings sheets.)



1 A-301 NORTH ELEVATION
SCALE : 1:100



2 A-301 EAST ELEVATION
SCALE : 1:100



3 A-301 SOUTH ELEVATION
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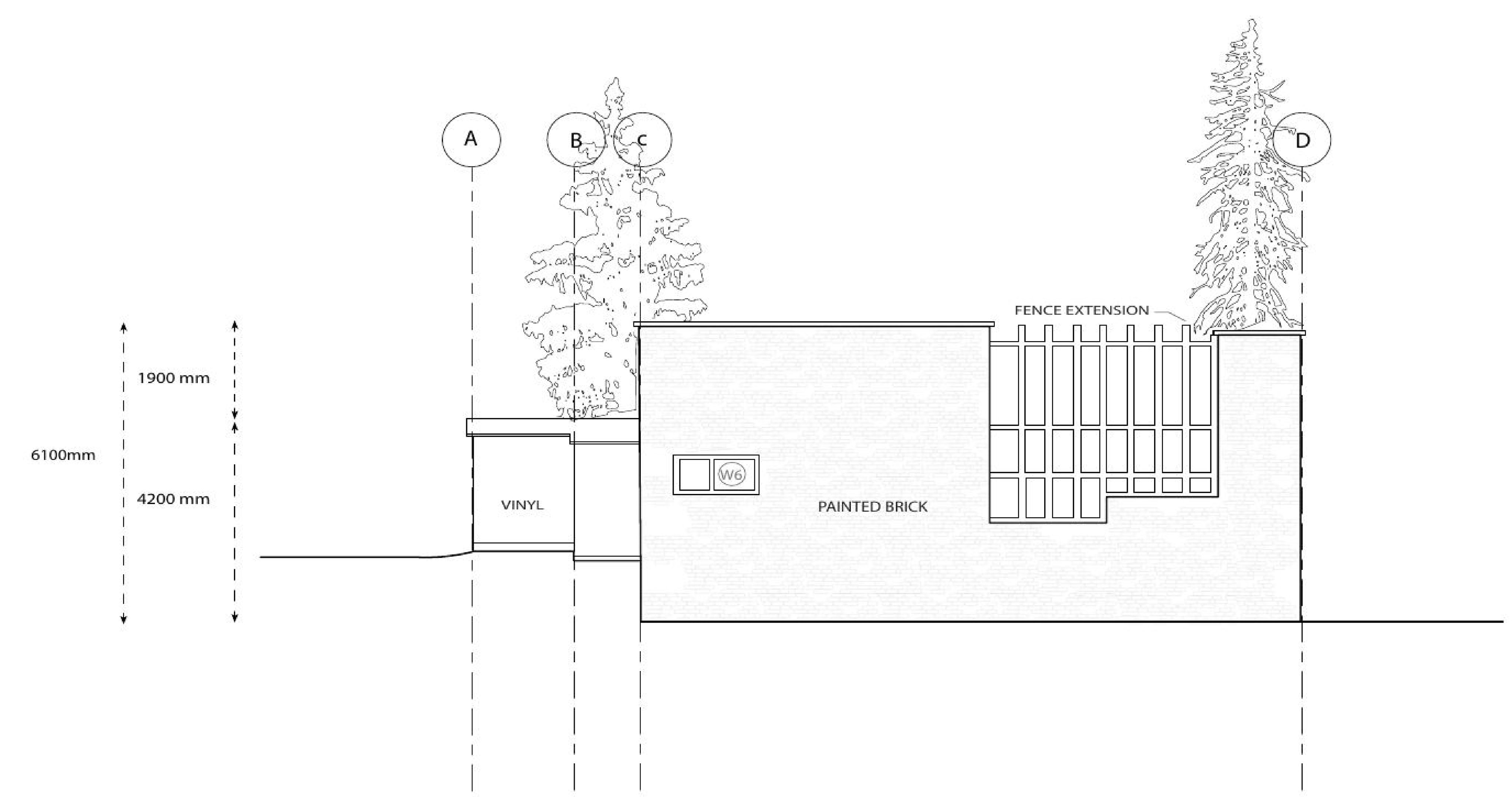
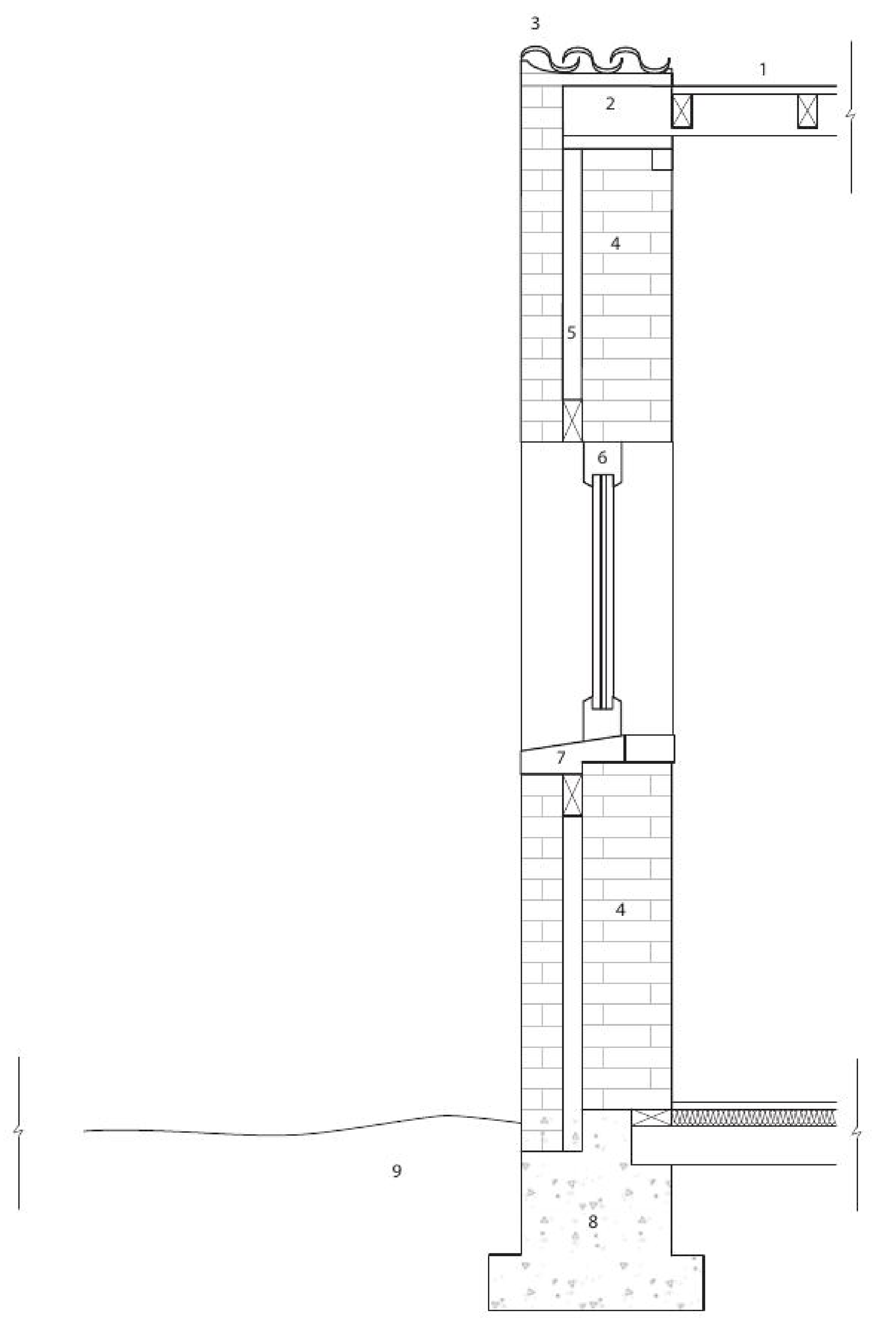


4 A-301 WEST ELEVATION
SCALE : 1:100

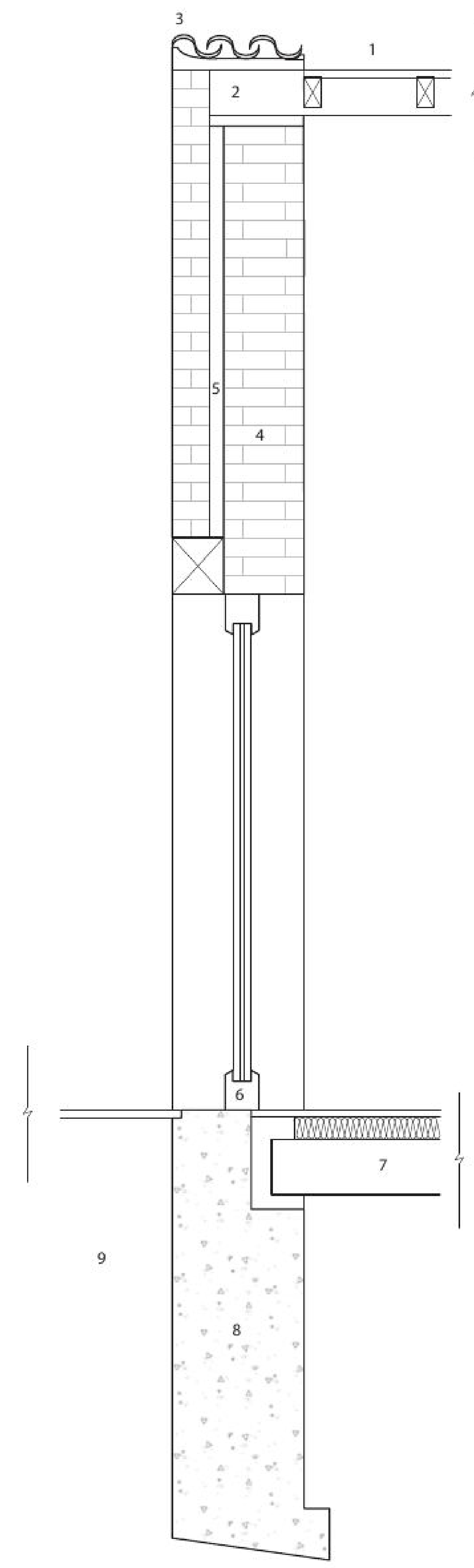
Carleton University Azrieli School of Architecture and Urbanism	EXPERIMENTAL HOUSE MELALAMMETIE,40900, FINLAND DATE OF CONSTRUCTION 1952-21954	CLIMATE AND ENVIRONMENTAL RETROFIT GROUP 7	SHEET NUMBER: A - 301 ELEVATIONS	TEAM MEMBER: NOORAIZ NOORANI	REVISION SCHEDULE: Nooraiz Noorani: 9/17/2025 - (Created line drawings from digital model) Nooraiz Noorani: 10/10/2025 - (adjusted line weights, added ground lines and elevation accuracy in accordance to TA feedback) Nooraiz Noorani: 10/11/2025 - (added annotations, dimensions and tags for clarity Nooraiz Noorani: 10/13/2025 - (Revised annotations, dimensions and tags) Nooraiz Noorani: 10/14/2025 - (Revised for consistency in Graphics across drawing sheets)



1 | A4

1 : 20 DETAIL WALL SECTION

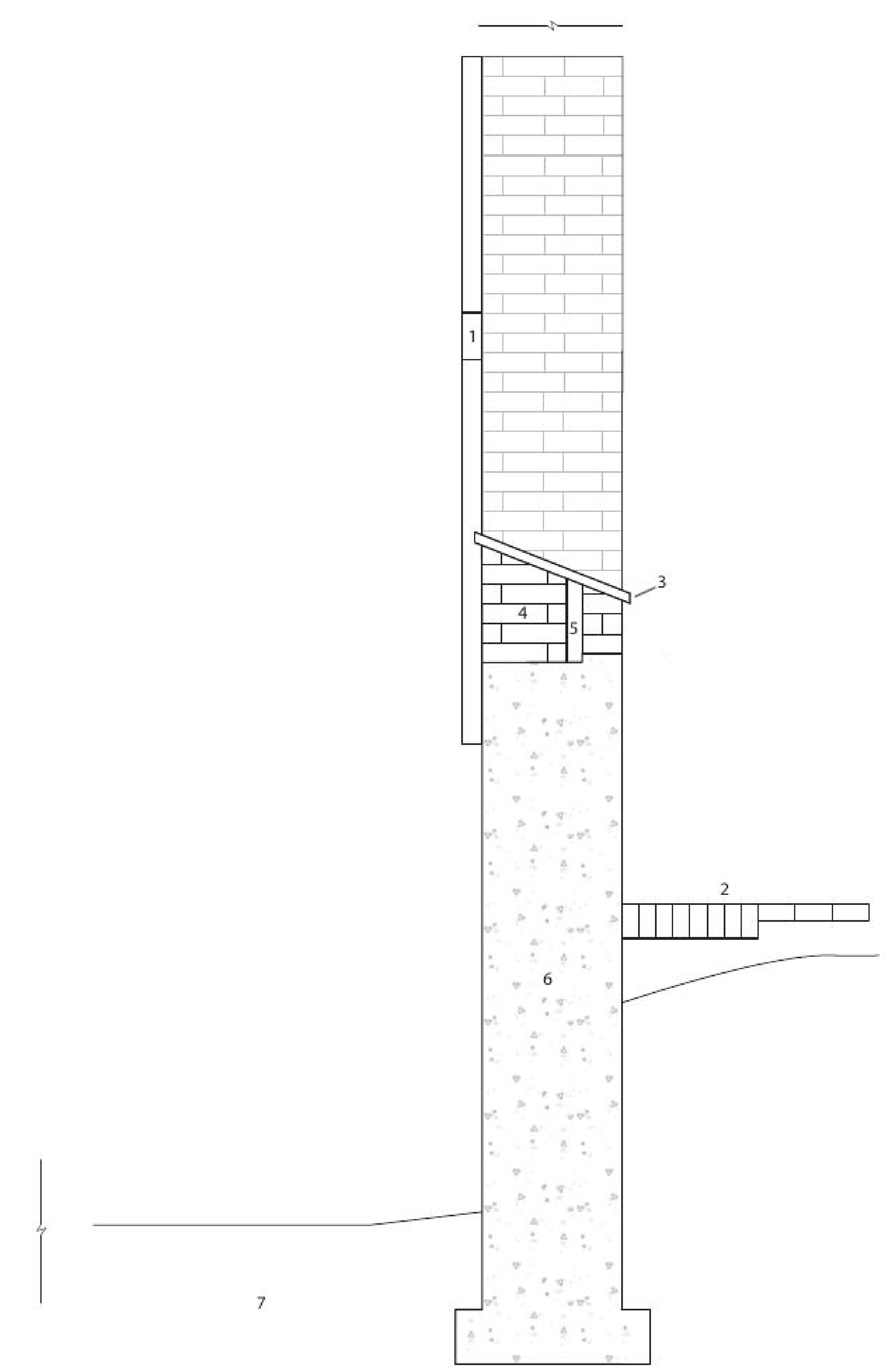
1 : ASPHALT
2 : WOOD BEAM
3 : CERAMIC SHINGLES
4 : BRICK
5 : AIR GAP
6 : WOOD WINDOW FRAME
7 : SILL
8 : CEMENT FOUNDATION
9 : GROUND



2 | A4

1 : 20 DETAIL WALL SECTION

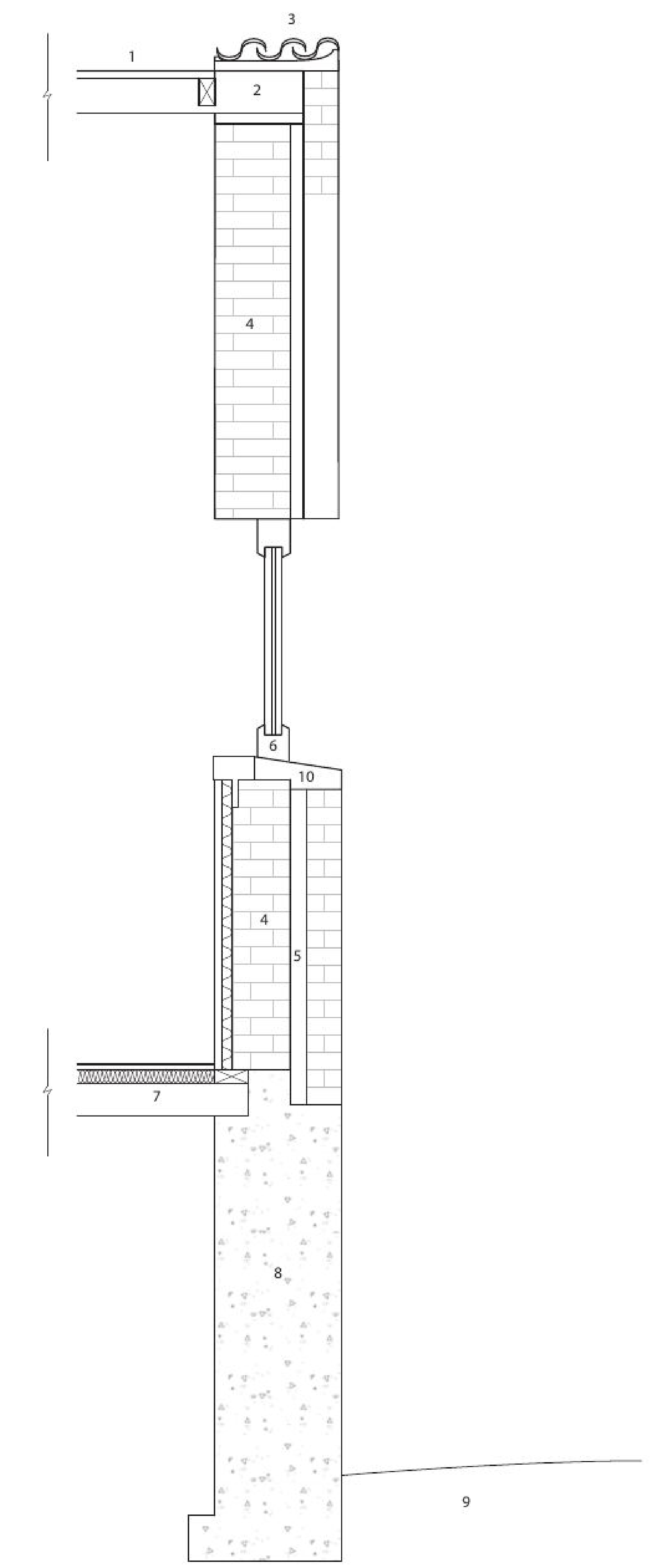
1 : ASPHALT
2 : WOOD BEAM
3 : CERAMIC SHINGLES
4 : BRICK
5 : AIR GAP
6 : WOOD WINDOW FRAME
7 : FLOOR JOIST
8 : CEMENT FOUNDATION
9 : GROUND



3 | A4

1 : 20 DETAIL WALL SECTION

1 : WOOD FENCEING
2 : BRICK PAVERS
3 : CERAMIC SHINGLES
4 : BRICK
5 : AIR GAP
6 : CEMENT FOUNDATION
7 : GROUND



4 | A4

1 : 20 DETAIL WALL SECTION

1 : ASPHALT
2 : WOOD BEAM
3 : CERAMIC SHINGLES
4 : BRICK
5 : AIR GAP
6 : WOOD WINDOW FRAME
7 : FLOOR JOIST
8 : CEMENT FOUNDATION
9 : GROUND
10 : SILL

Carleton University
Azrieli School of
Architecture and Urbanism

EXPERIMENTAL HOUSE
MELALAMMETIE,40900,
FINLAND

DATE OF CONSTRUCTION
1952-21954

CLIMATE AND ENVIRONMENTAL RETROFIT

GROUP 7

SHEET NUMBER:

A - 401

WALL SECTIONS

TEAM MEMBER:

OWEN KEEPING

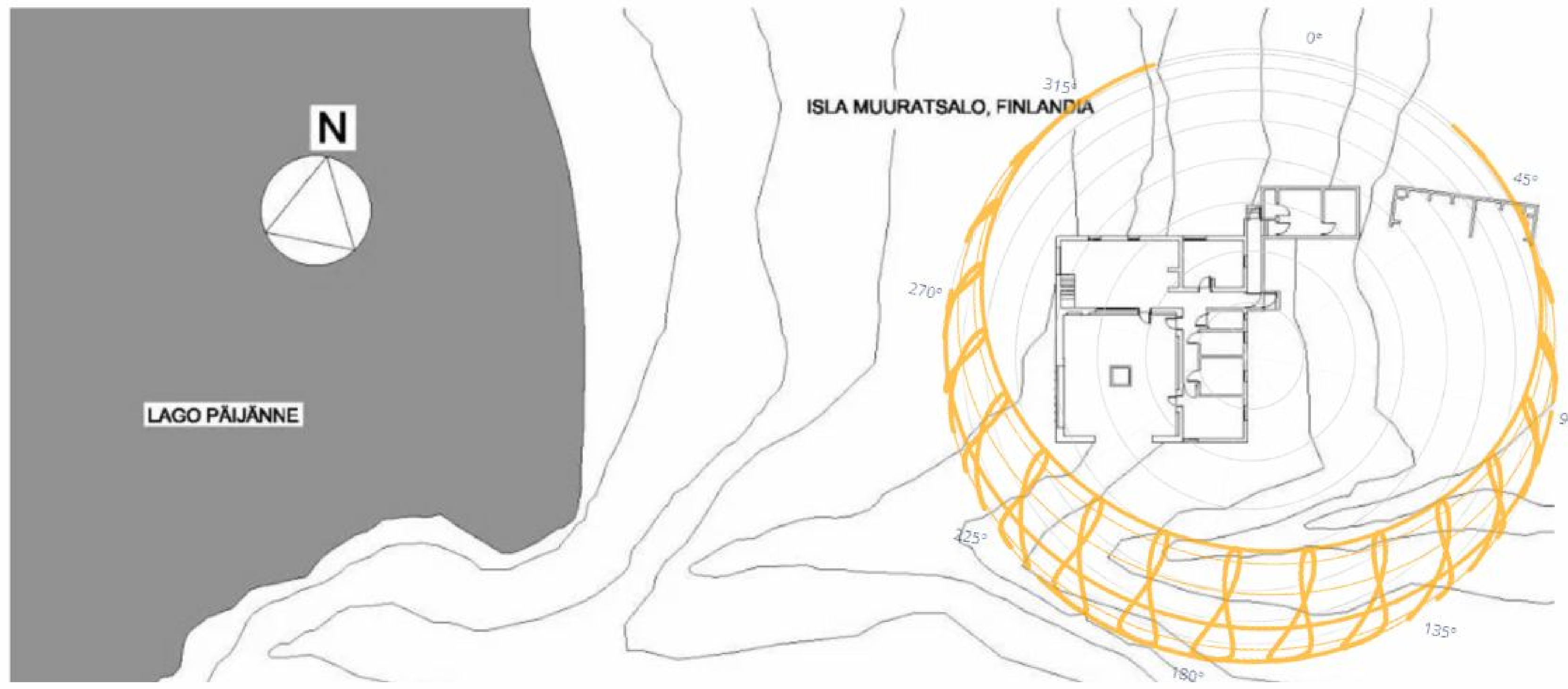
REVISION SCHEDULE:

Nooraiz Noorani 9/24/2025 - Created draft wall detail for TA feedback and group understanding
Colin Murray: 10/01/2025 - 10/06/2025 (Created preliminary wall detail drawing)
Colin Murray: 10/06/2025 - (Modified preliminary wall detail drawing according to group member feedback)
Owen Keeping: 10/06/2025 - (Collaborated to improve first wall detail.)
Owen Keeping: 10/09/2025 - (Worked form model and sections to create the remaining detailed wall sections.)
Owen Keeping: 10/11/2025 - (Refined construction layers and connection details for accuracy.)
Owen Keeping: 10/14/2025 - (Added annotations.)
Owen Keeping: 10/14/2025 - (Revised for consistency in Graphics across drawings sheets.)

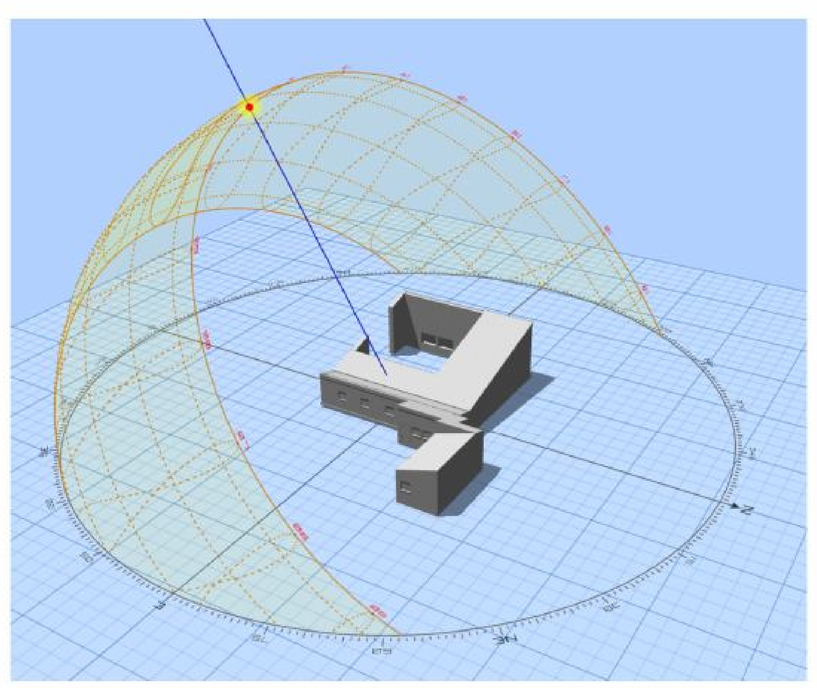
Solar Analysis

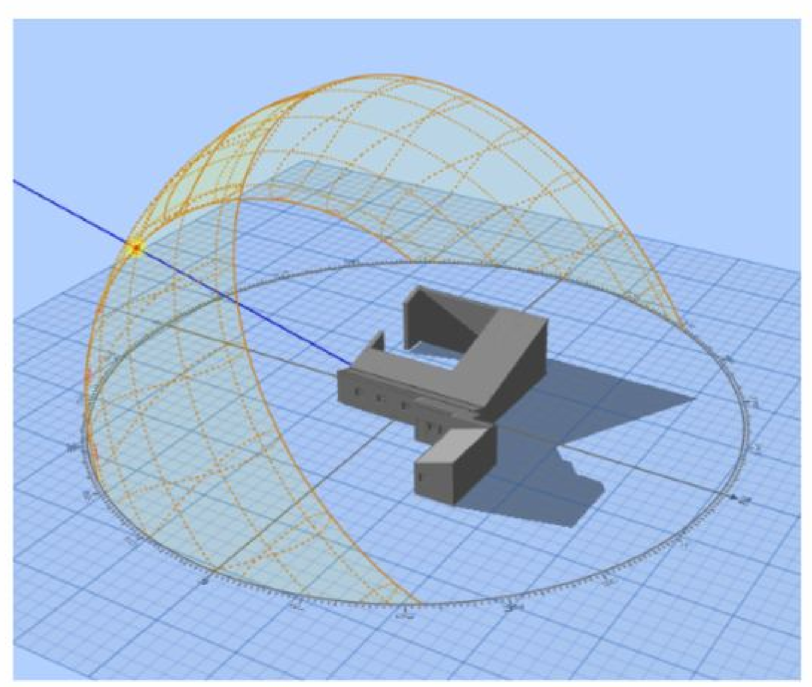
Current location: Jyvaskyla, Finland
Proposed location: Portland Oregon

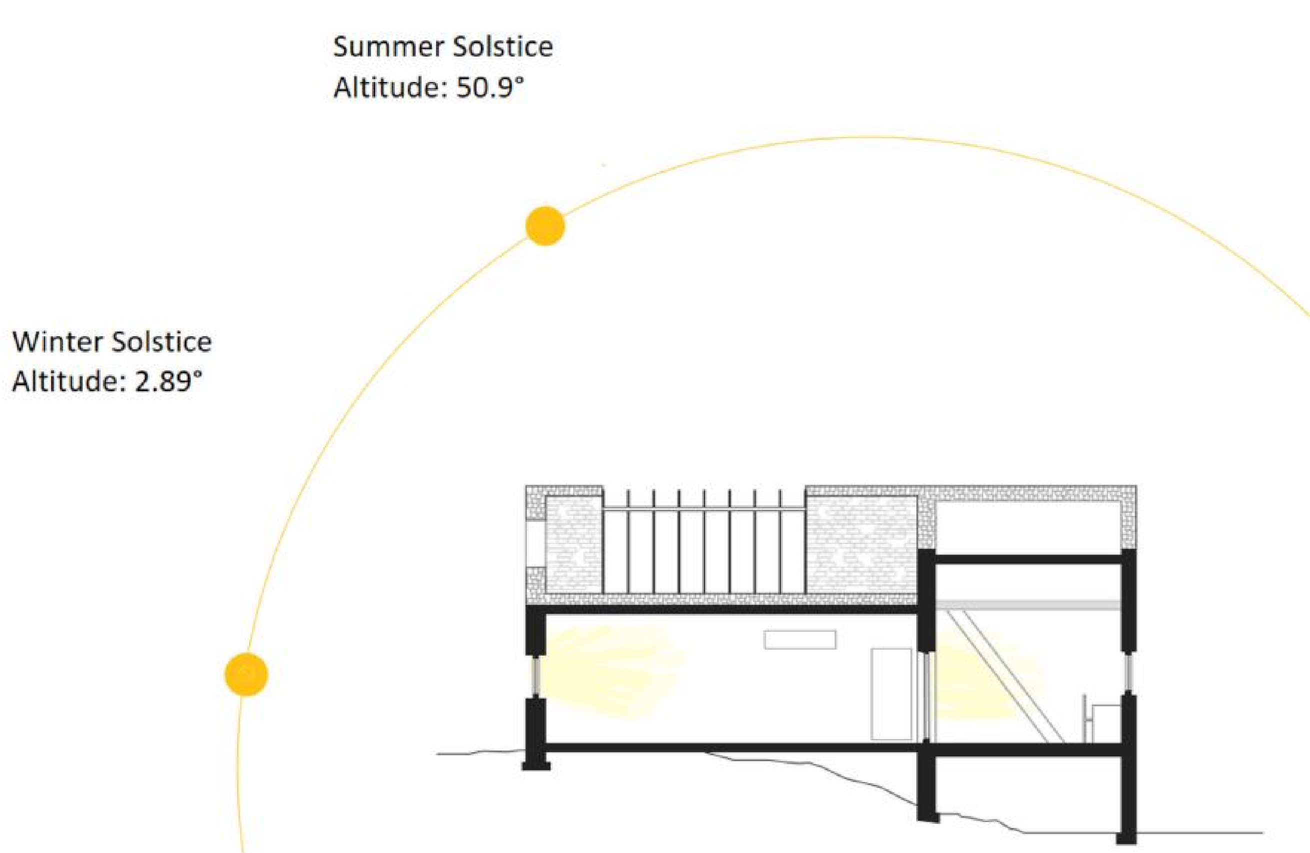
Summary of current solar climate



The Experimental House in Finland is oriented with its courtyard opening nearly true south, allowing sunlight to penetrate the large patio doors along the south wall. As the courtyard forms the central focus and receives the most sunlight year-round, maintaining a south-facing orientation would be essential when selecting a new site. This house was designed as a summer home and therefore is not usable for over half of the year in it's current location. When deciding where to move the house, we wanted a location that would increase the usability of the home while keeping in mind the limitations on the design such as the lack of insulation for colder days

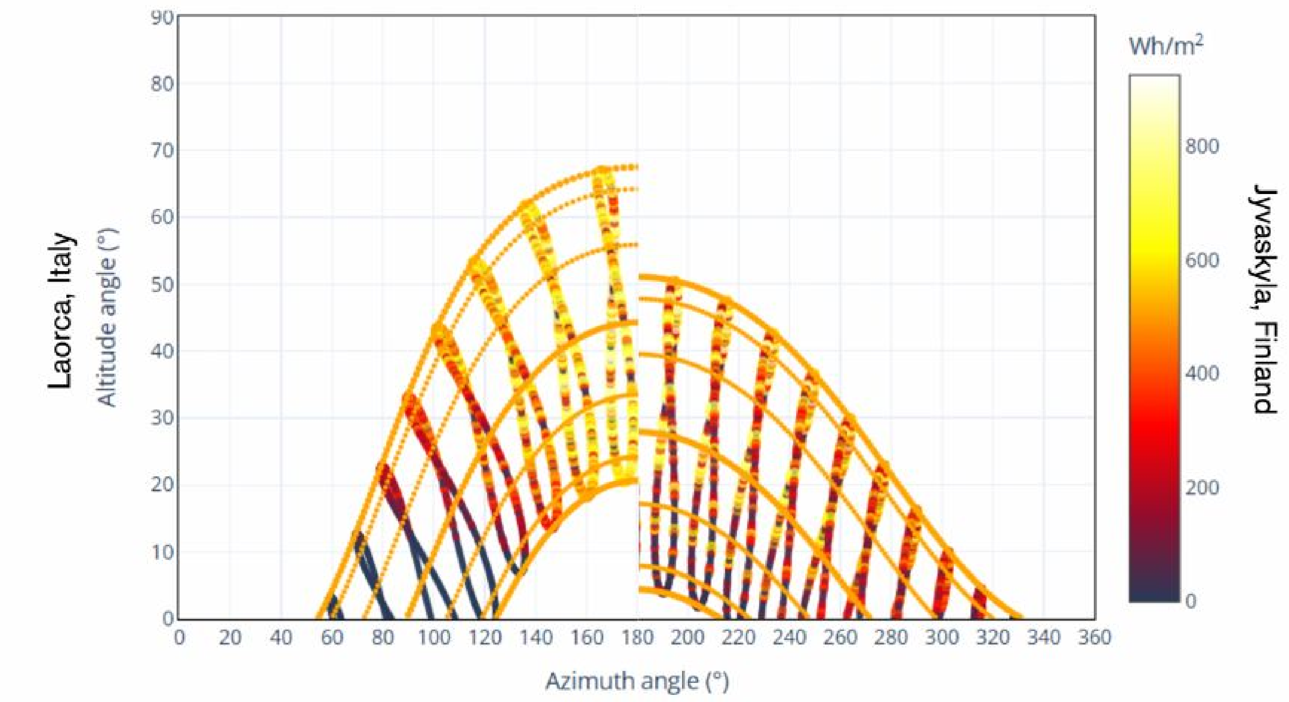






Considered Locations

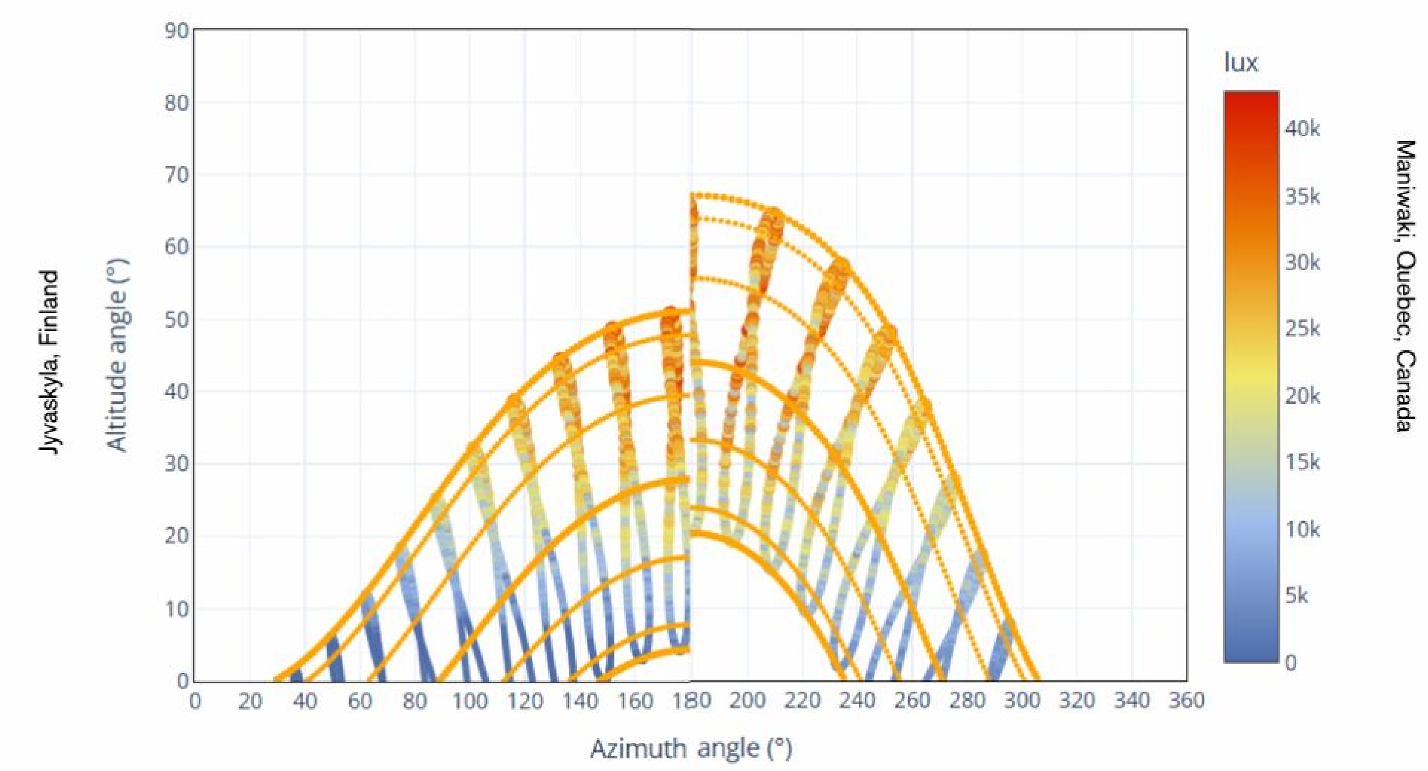
Laorca, Lecco, Italy

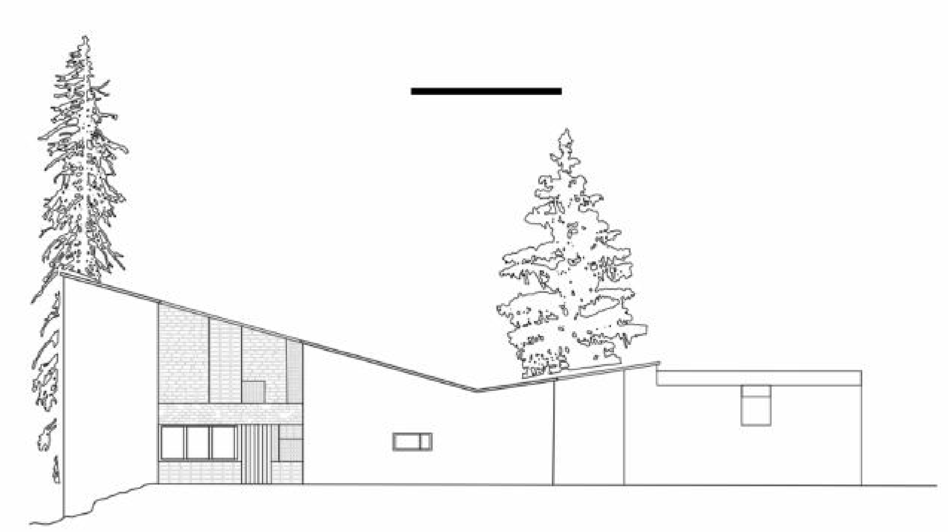


Finland's annual horizontal solar radiation averages only 878.7 kWh/m² due to frequent cloud cover, limiting natural daylight and the effective use of the courtyard. Relocating the house to Laorca, where solar radiation is nearly 50% higher, would enhance light and usability but introduce potential issues of summer overheating. Therefore we decided not to go with Italy due to too much solar gain and hotter climate that the house is not designed for.

Maniwaki, Quebec, Canada

Comparing the Climates of Finland and Central Quebec have numerous similarities leading to Maniwaki becoming a potential new location. Finland at most has a 59% chance of clear skies while Quebec has a 65%. Relocation will obtain higher illuminance and solar altitude. This indicates less need for artificial lighting and more access to daylight advantages. While this location would increase the solar potential of the house, other factors such as colder winters would still make this home unusable for most of the year. Therefore moving the house to Quebec is not the most practical site for the Experimental house

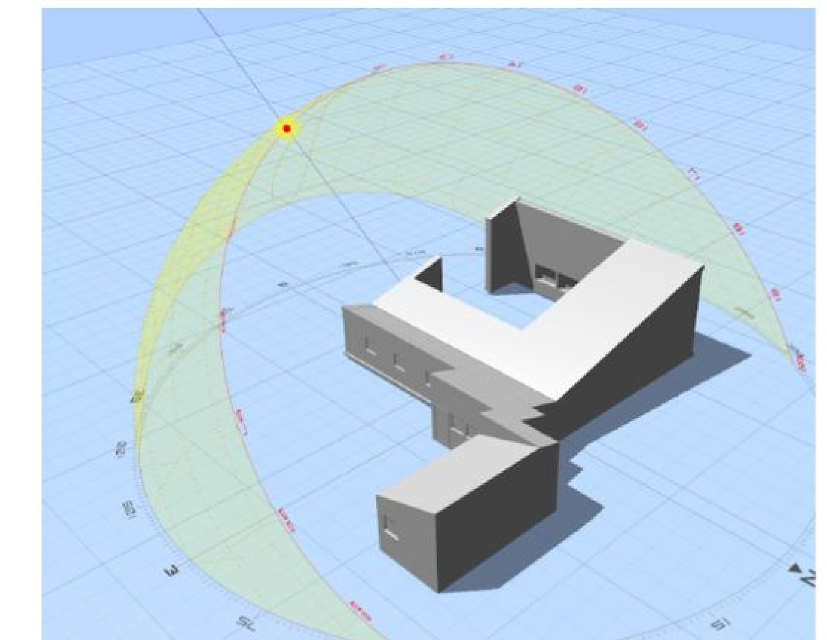


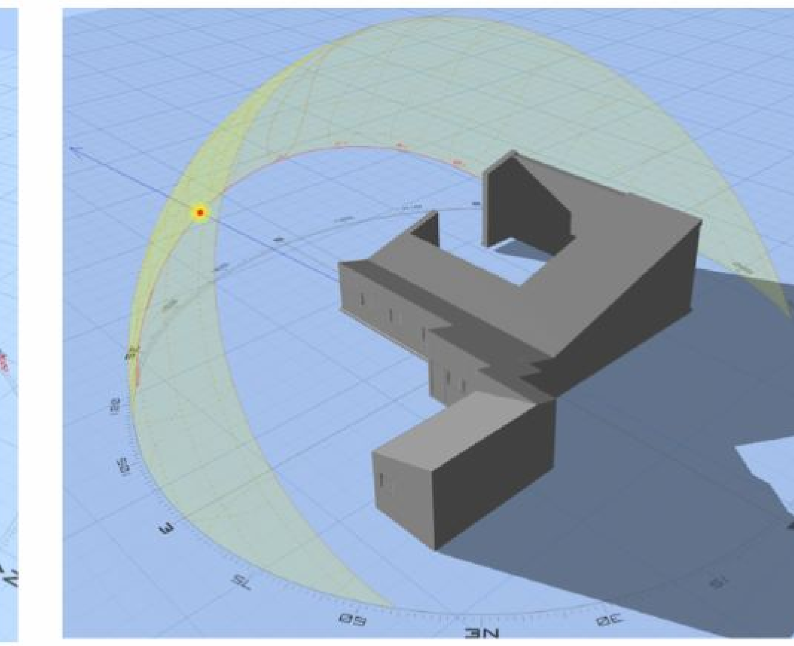


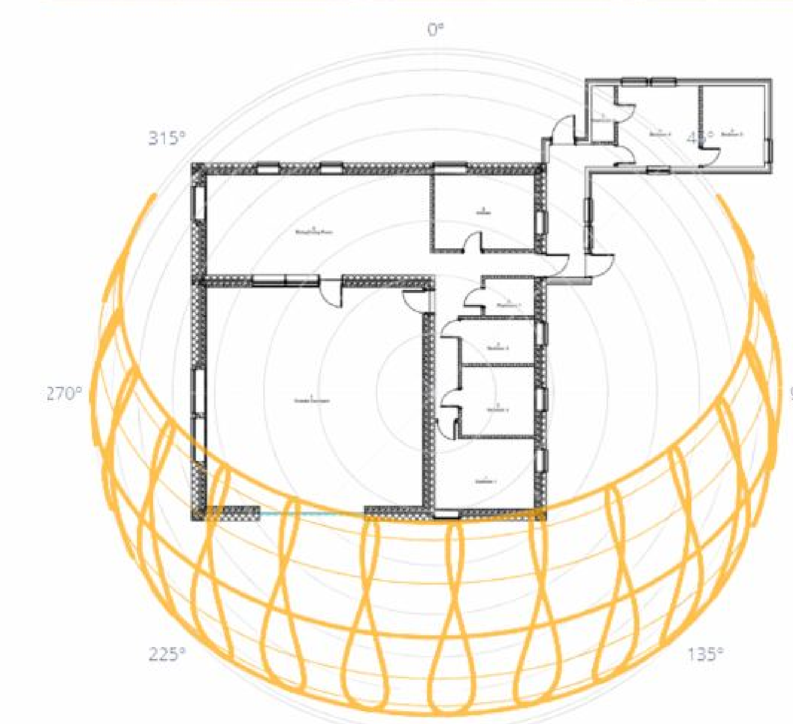
With keeping the original orientation of the house in any of the proposed locations , we maximize the optimal use of the courtyard but limit the amount of natural light that will enter the home due to the small window to wall ratio of 6.3% on the south elevation. This ratio is well below the typical amount and is not ideal for the house. This is one draw back of keeping the orientation of the house.

Final Site
Outside Portland, Oregon, USA

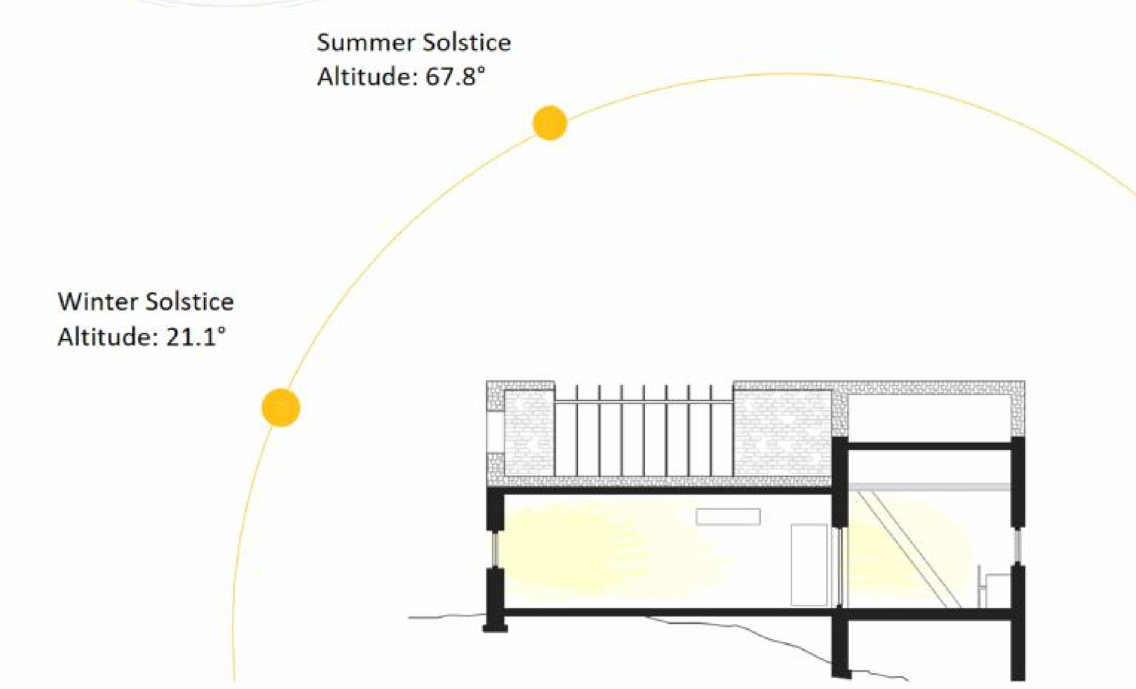
Jyvaskyla lies at 62° N, while Portland's lower latitude of 45.5° N results in higher solar elevations, reduced seasonal extremes, and longer daylight hours throughout the year. As shown in the sun path diagrams below, the higher sun altitude would greatly benefit the natural light entering the home during the winter months.



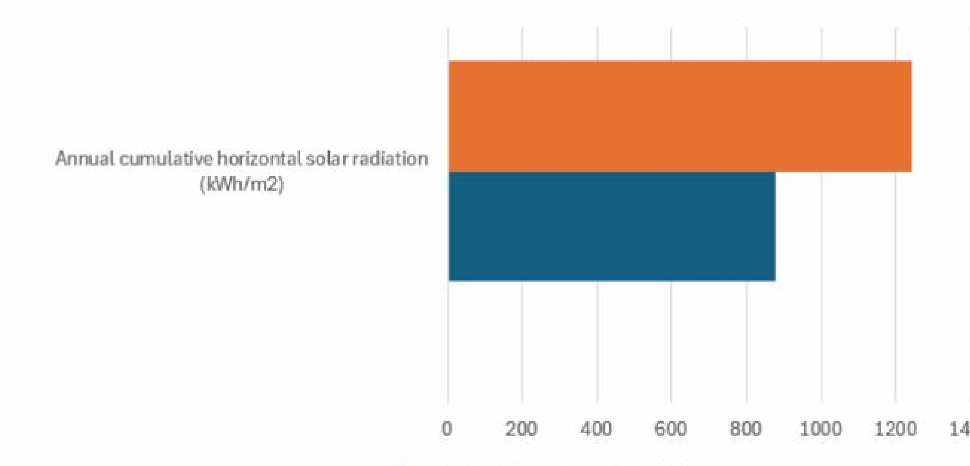




Aalto designed the house as a summer home, limiting its use to a few months each year. Relocating the home to a warmer, sunnier climate would increase the usability while maintaining occupant comfort. In the new location, we would keep the house positioned the same way Aalto intended so the courtyard receives the most amount of sun. The west side of the courtyard has a large opening that allows the afternoon sun to come through and reach the house.



Overall, moving the house to outside of Portland, Oregon has the best impact on solar potential on the house mainly due to the lower altitude. We would keep the original orientation of the house and relocate to a similar terrain as Finland. Outside of downtown Portland is various rocky and lush forest that would maintain the desirable climate that Aalto loved in the original site.



Carleton University
Azrieli School of
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EXPERIMENTAL HOUSE
MELALAMMETIE,40900,
FINLAND

DATE OF CONSTRUCTION
1952-21954

CLIMATE AND ENVIRONMENTAL RETROFIT

GROUP 7

SHEET NUMBER:
A - 501
SOLAR
ANALYSIS

TEAM MEMBER:
KAITLIN GIBSON

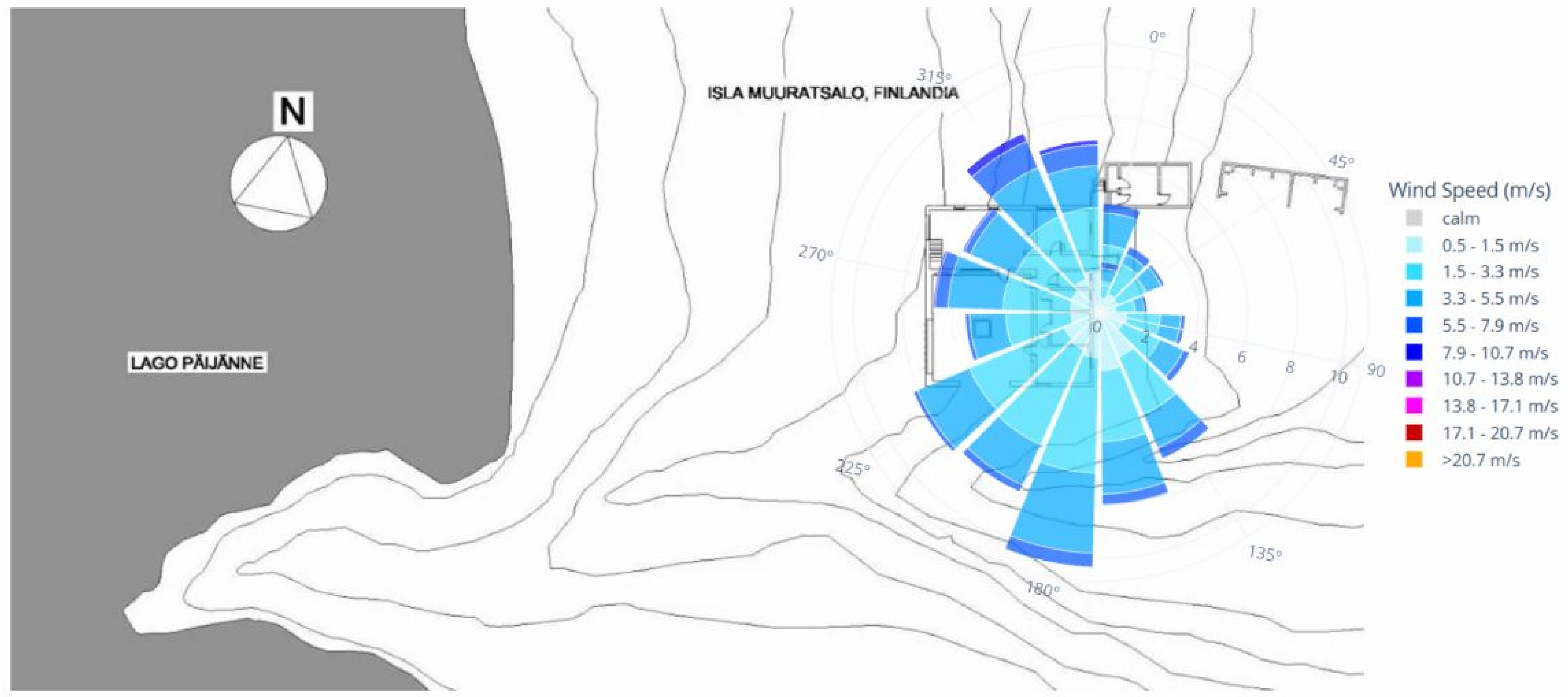
REVISION SCHEDULE:

NOORAIZ NOORANI	10/01/2025	CREATED 3D GRAPHIS USING ANDREW MARSH TOOLS
KAITLIN GIBSON	10/01/2025	CREATED OVERALL FORMAT AND LAYOUT FOR ANALYSIS
KAITLIN GIBSON	10/01/2025	CONDUCTED SOLAR ANALYSIS USING CLIMA TOOL
KAITLIN GIBSON	10/05/2025	ANNOTATED GRAPHS AND CHARTS
MANREET ATWAL	10/05/2025	ANALYSIS ON QUEBEC LOCATION
KAITLIN GIBSON	10/12/2025	ADDED MORE VISUAL DIAGRAMS FOLLOWING PIN-UP

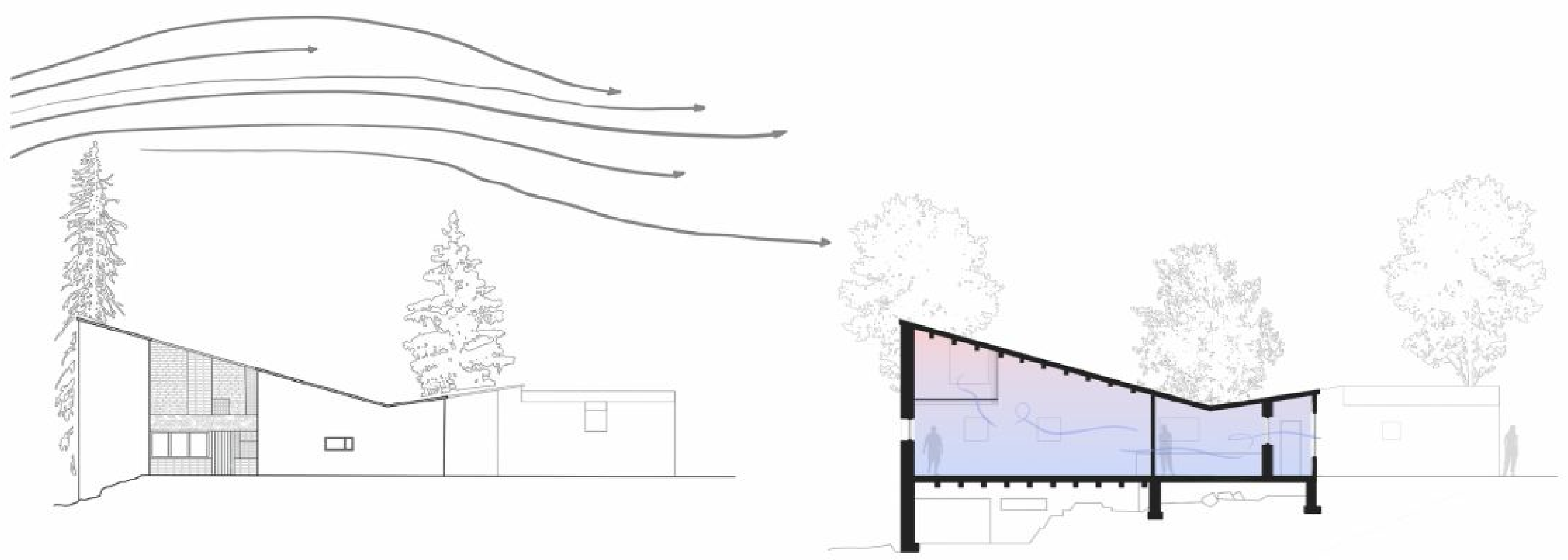
Wind Analysis

Current location: Jyvaskyla, Finland
Proposed location: Portland Oregon

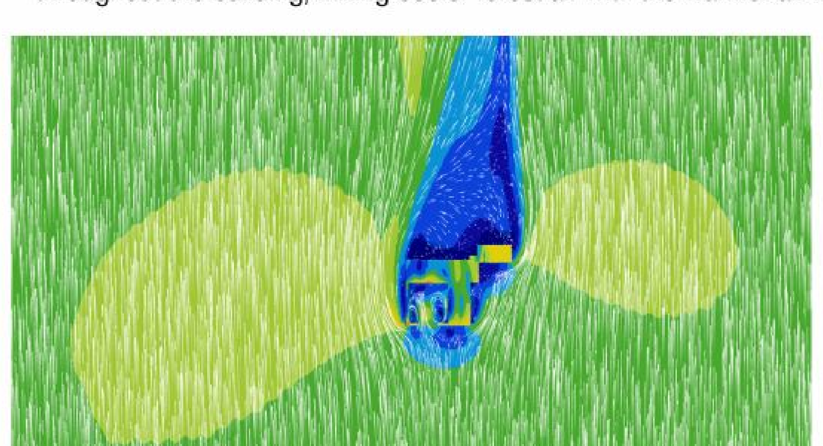
Summary of Current Solar Climate



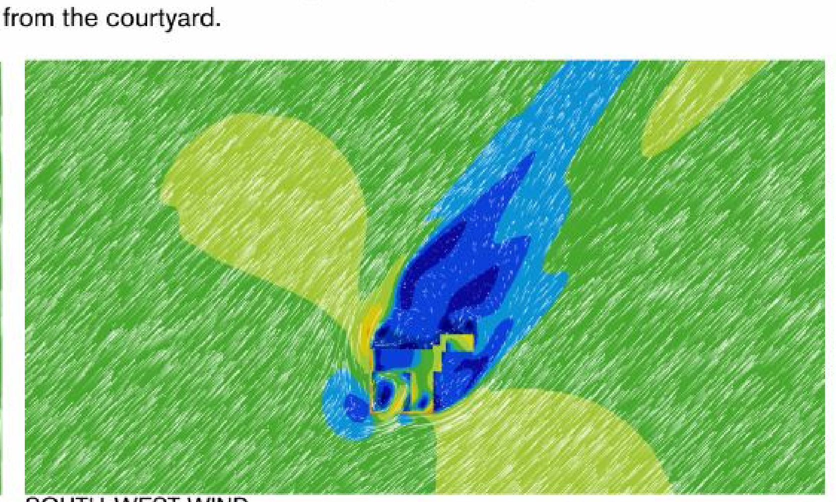
The Experimental House experiences irregular but moderate winds typical of the region's northern climate. The area is subject to variable airflow influenced by the surrounding lake and forested terrain. The diagram illustrates winds primarily originating from the south and southwest, with additional flows from the west and northwest. Wind speeds are generally moderate, ranging between 1.5 m/s and 7.9 m/s, with occasional stronger gusts exceeding 10 m/s.



Aalto's architectural design effectively responds to its natural surroundings. The house is situated in a pine forest and is positioned behind a rocky outcrop, which shields it from cold winds. This design reduces turbulence, allowing air to circulate naturally. The living room wing allows warm air to rise and escape through high windows, while cooler air enters through lower openings, creating an upward airflow that keeps the space comfortable. Additionally, a windowed corridor between the kitchen and the guest apartment helps air circulate throughout the building, mixing cooler forest air with the warmer air from the courtyard.



SOUTH WIND



SOUTH-WEST WIND

Wind analysis diagrams illustrate how the building form interacts with prevailing winds. Under both south and southwest wind conditions, the highest wind speeds (7–10 m/s, shown in orange to red) are diverted above and around the building, while areas immediately surrounding the house remain within the calmer 0–3 m/s range (blue to green). These sheltered zones promote stable, gentle airflow that naturally circulates through the openings. Together, these features enable steady, passive ventilation without the need for mechanical systems, showcasing Aalto's emphasis on harnessing natural airflow to maintain a comfortable and balanced home environment.

Considered Locations

Laorca, Lecco, Italy

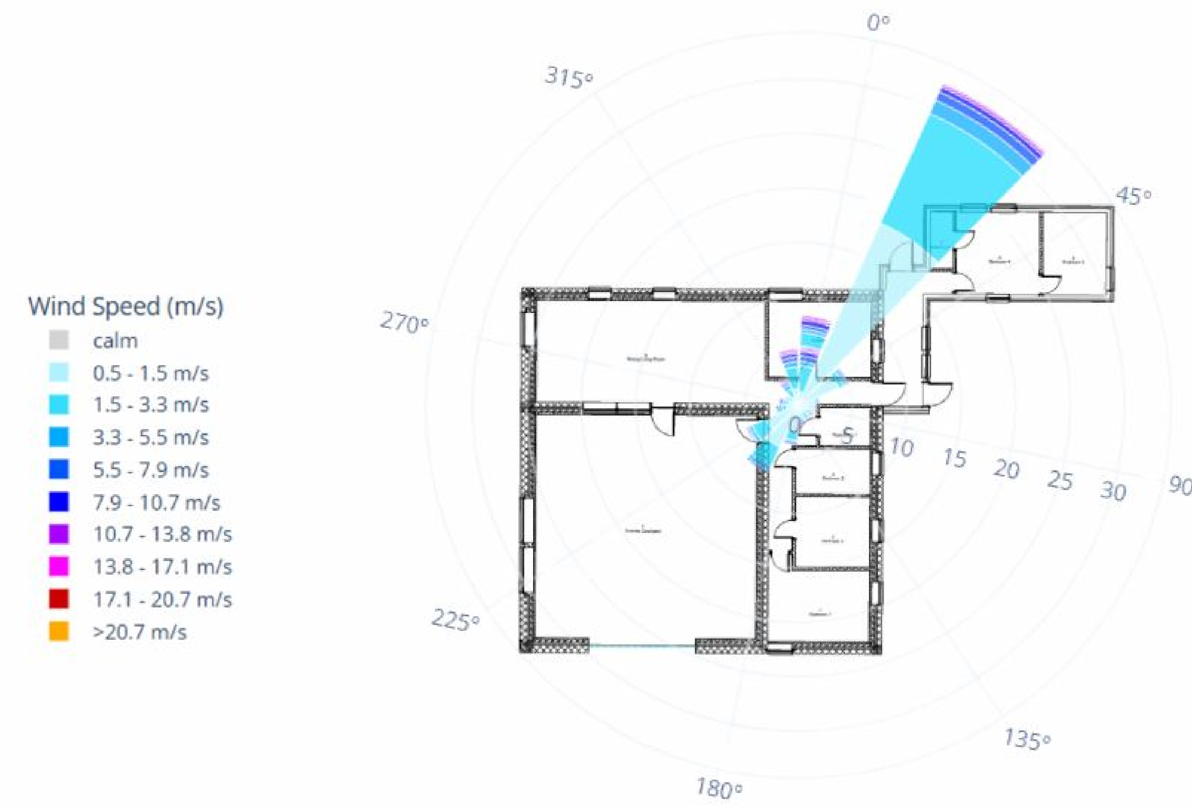
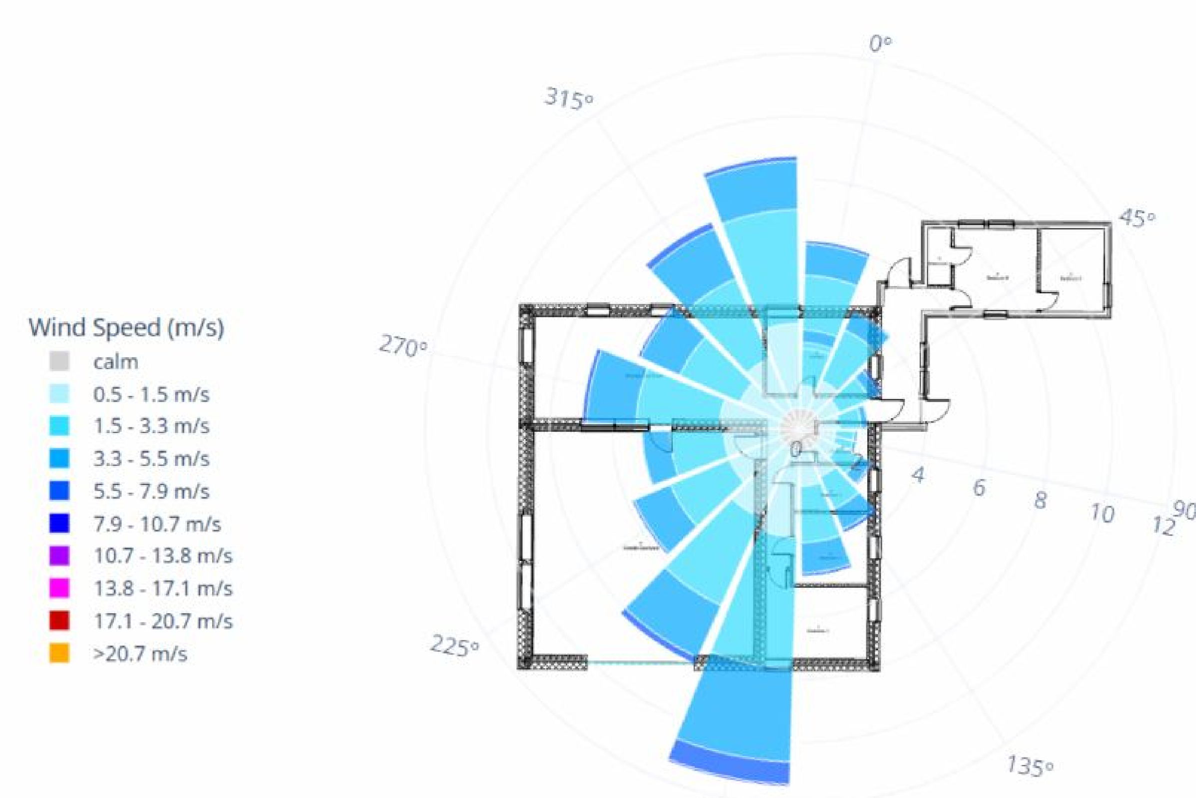


Diagram showcases average wind direction and intensity the building would receive in Laorca, lecco, Italy.

This location has stronger and more consistent winds than Muuratsalo, where the winds mainly blow from the north to northeast. The predictable wind direction in Laorca allows for better natural ventilation. In contrast, Muuratsalo's varying winds require a more balanced approach to airflow and shelter. However, in Muuratsalo, Aalto utilized the island's landscape and trees to shield the house while still allowing gentle air to circulate through the courtyard, thereby creating a calm microclimate. A site in Laorca would experience more intense winds, which would require additional protection on the north-facing walls and might limit the open spaces in the courtyards.

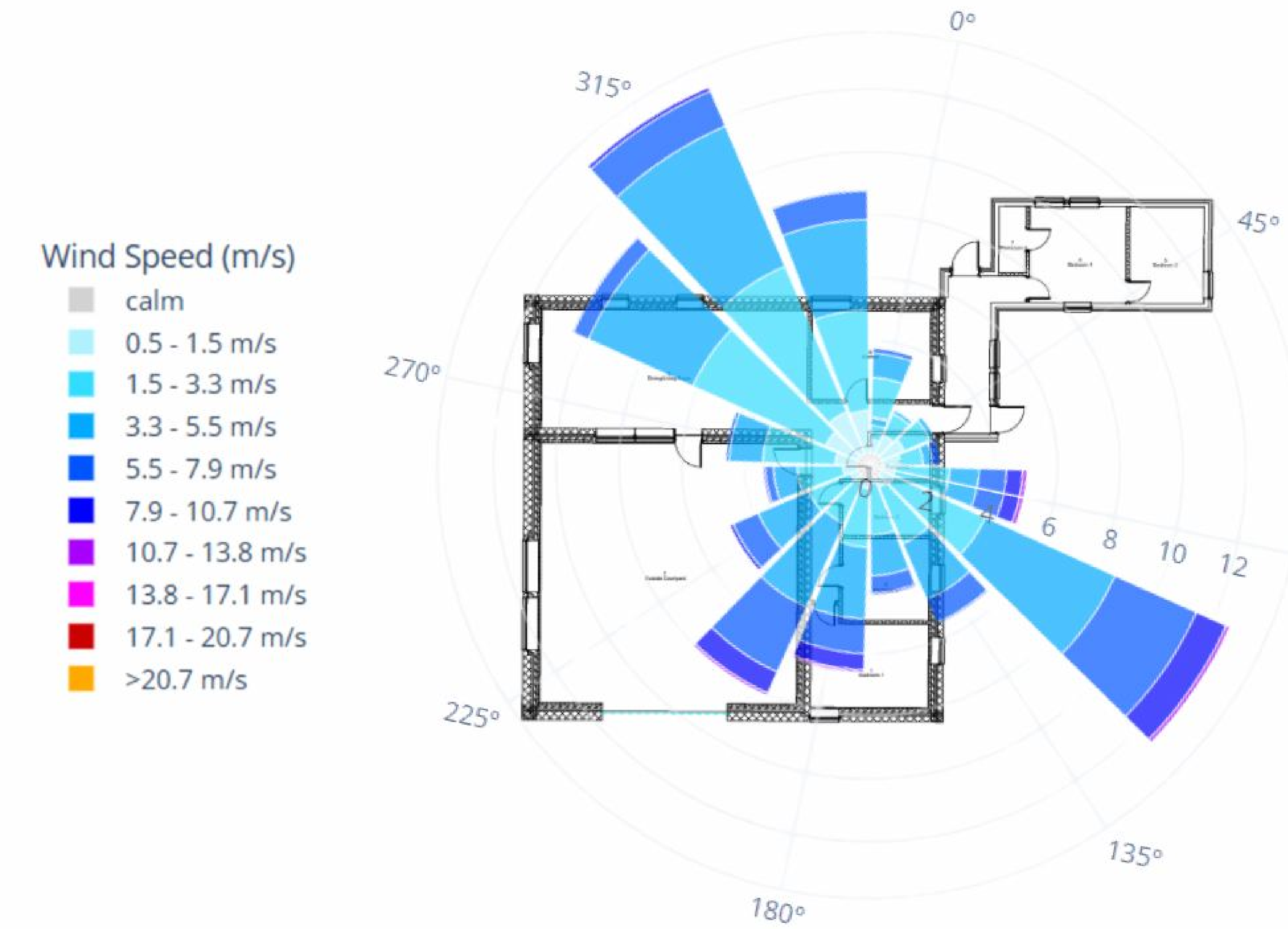
Maniwaki, Quebec, Canada



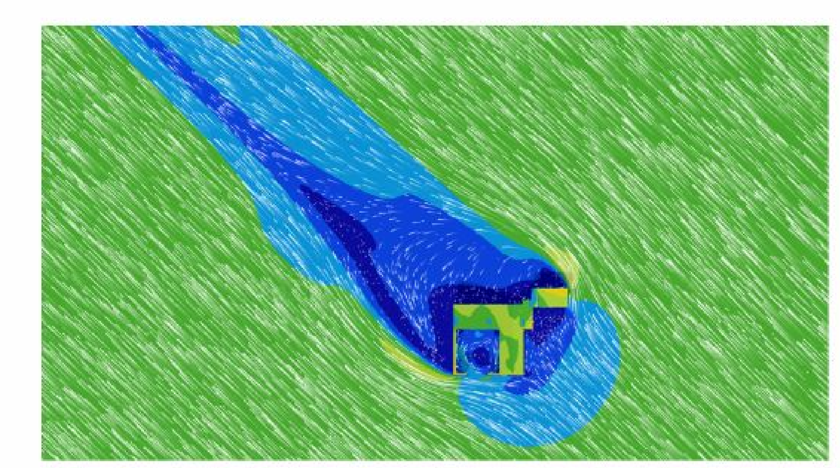
In Maniwaki, Quebec, prevailing winds primarily come from the south and southwest, with moderate speeds ranging from 1.5 to 7.9 m/s, creating a consistent and directional airflow influenced by the region's open valleys and forested surroundings. Compared to Muuratsalo, Maniwaki's winds are steadier and more exposed. If the Experimental House were placed here, its courtyard and openings could be oriented to capture these southerly breezes for natural ventilation, while screen walls would be needed to recreate the wind protection provided by the original island setting.

Final Site

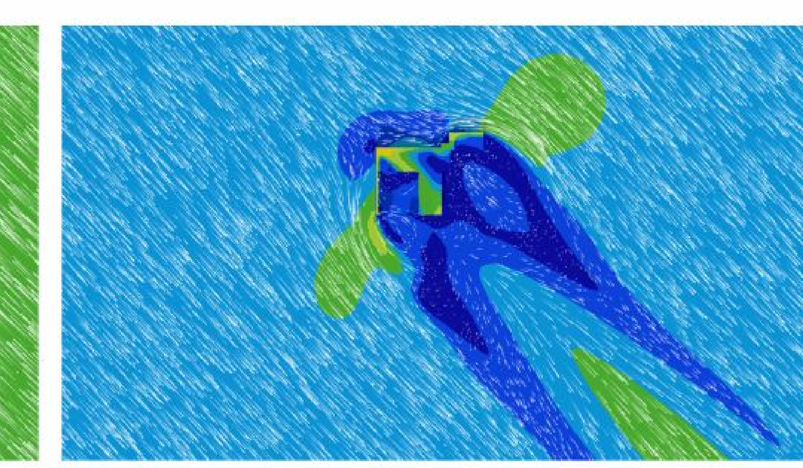
Outside Portland, Oregon, USA



In Portland, Oregon, the dominant winds mainly originate from the northwest and southeast, with minor contributions from the east and southwest, with speeds ranging from moderate to high (3-10 m/s), resulting in a consistent wind pattern. Under southeast wind conditions, the airflow is channeled smoothly along the site, with reduced turbulence zones (0–3 m/s, shown in blue and green) forming around the structure, creating calm and comfortable outdoor areas. Meanwhile, northwest winds produce a similar pattern, with wind flow directed around the building mass and minimal high-speed zones near the walls, indicating effective wind deflection and stable ventilation potential.



SOUTH-EAST WIND



NORTH-WEST WIND

Portland presents a more reliable and temperate wind climate, making it advantageous for passive cooling and airflow design. This location was selected as the preferred site because its steady winds and potential for natural ventilation align with the project's environmental objectives. Although these conditions differ from the original site's wind patterns, Aalto's passive design principles remain effective in adapting to varying climates and wind directions.

Carleton University Azrieli School of Architecture and Urbanism	EXPERIMENTAL HOUSE MELALAMMETIE,40900, FINLAND DATE OF CONSTRUCTION 1952-21954	CLIMATE AND ENVIRONMENTAL RETROFIT GROUP 7	SHEET NUMBER: A - 601 WIND ANALYSIS	TEAM MEMBER: SOFIA CANAS MORALES

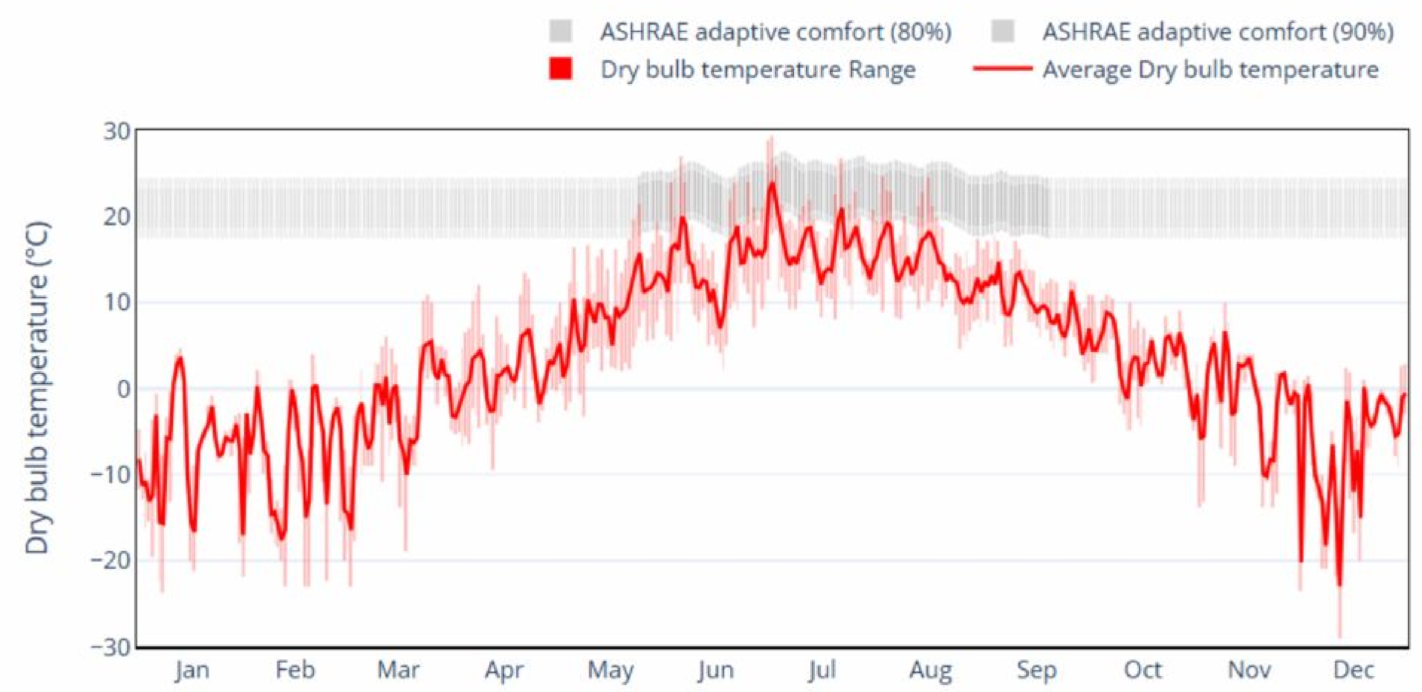
REVISION SCHEDULE:		
NOORAIZ NOORANI	10/01/2025	CREATED 3D GRAPHS DISPLAYING WIND ANALYSIS
SOFIA CANAS M	10/07/2025	FORMATTED THE SHEET FOR ANALYSIS
SOFIA CANAS M	10/07/2025	CONDUCTED WIND ANALYSIS USING CLIMA TOOL
SOFIA CANAS M	10/07/2025	ANNOTATED WIND ANALYSIS GRAPHS FOR EACH LOCATION
SOFIA CANAS M	10/14/2025	SLIGHTLY REFORMATTED THE LAYOUT ACCORDING TO FEEDBACK
SOFIA CANAS M	10/14/2025	MODIFIED THE DIAGRAMS ACCORDING TO FEEDBACK
SOFIA CANAS M	10/14/2025	MODIFIED THE ANNOTATIONS ACCORDING TO FEEDBACK

Thermal Analysis

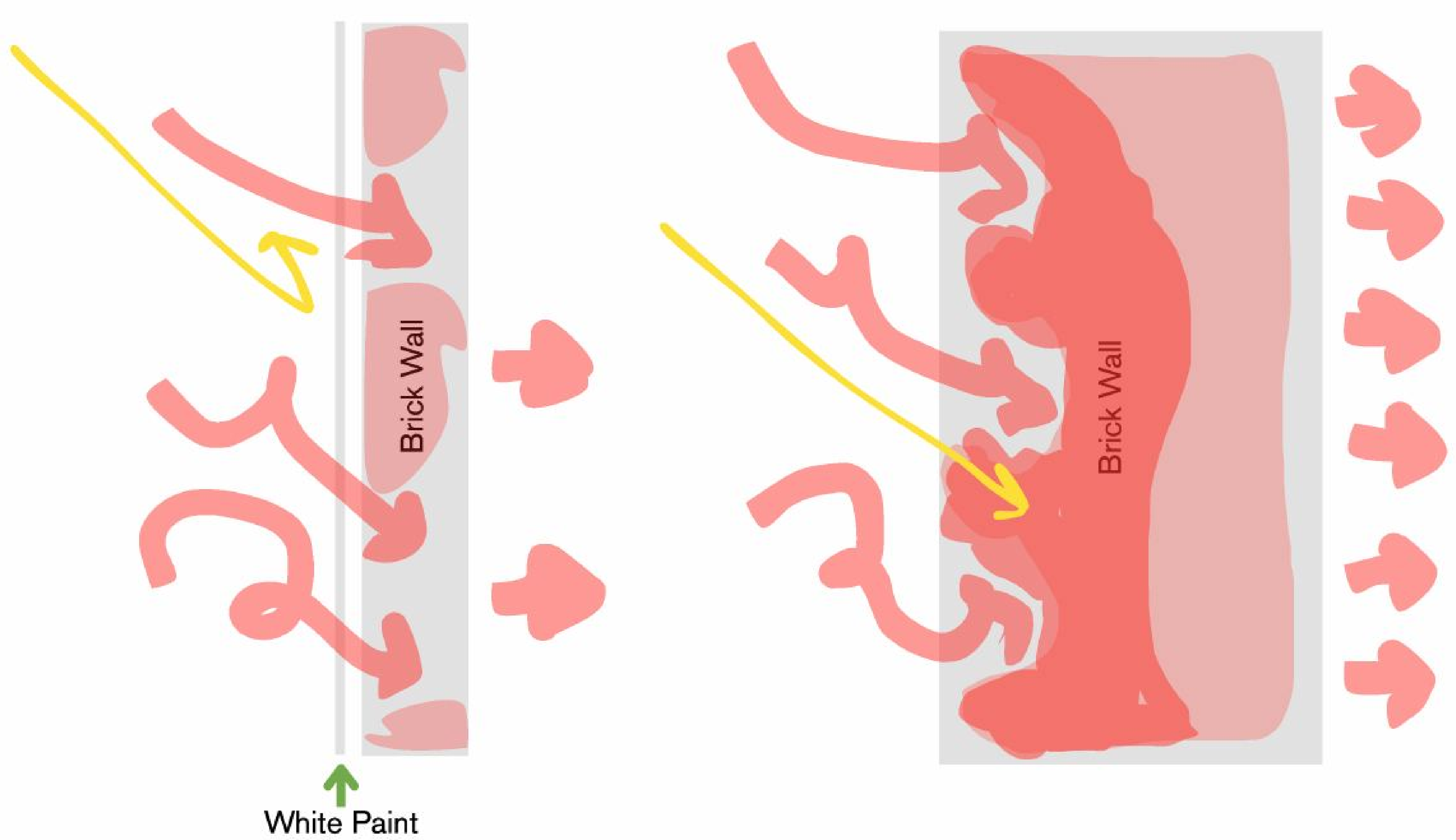
Current location: Jyvaskyla, Finland

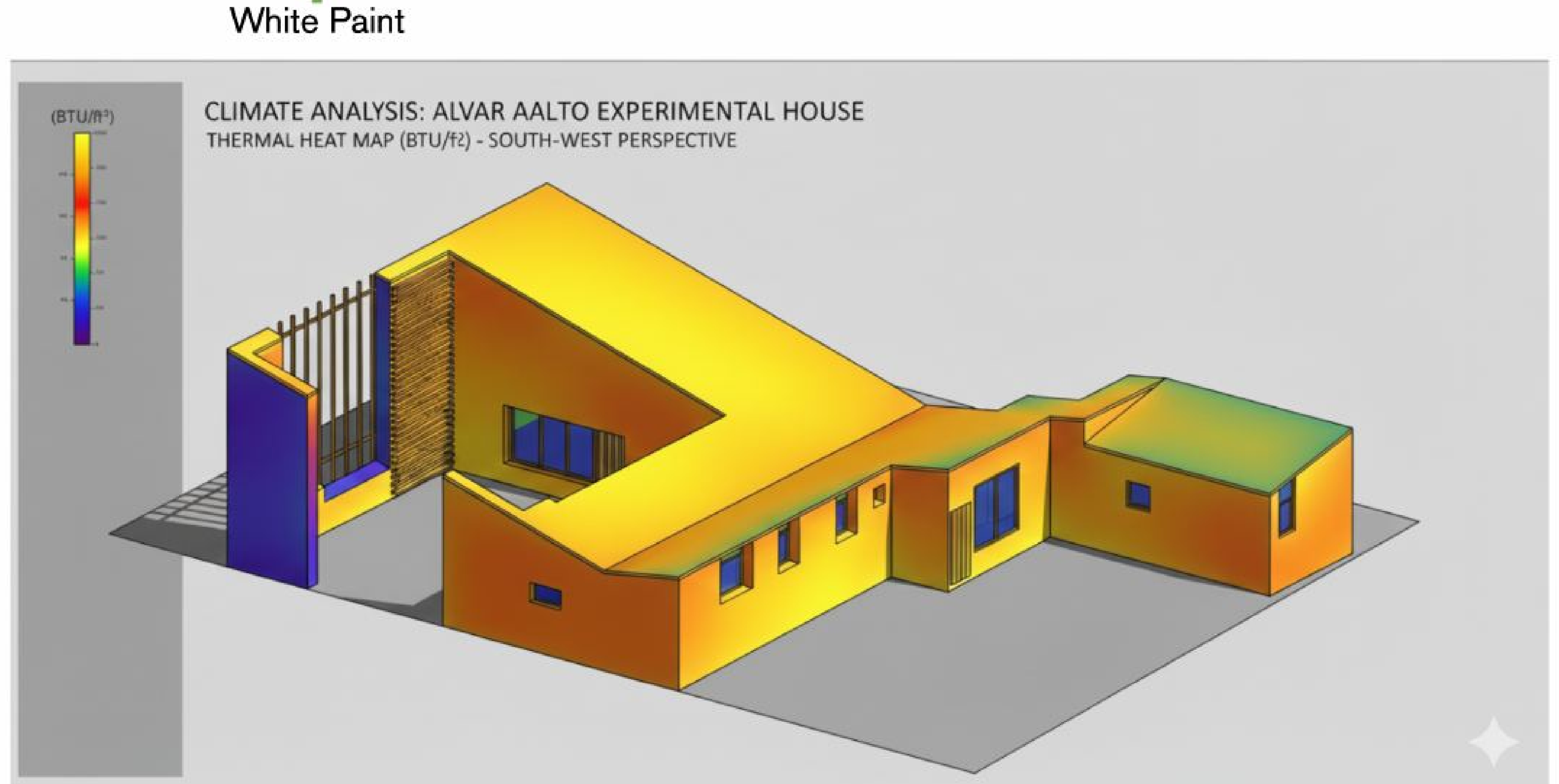
Proposed location: Portland Oregon

Summary of Current Thermal climate



Currently the Experimental House remains unoccupied. The structure itself has no insolation or way to combat heat loss as it was designed originally as a summer home. The current climate makes it impossible for occupants in most months. The house is however, design to block the cool lake breeze in the west and does make use of the solar for heating. The building is constructed of mostly brick, which is a thermal massing material. Its function is less efficient do to the counter productive white paint and thin walls.





Considered Locations

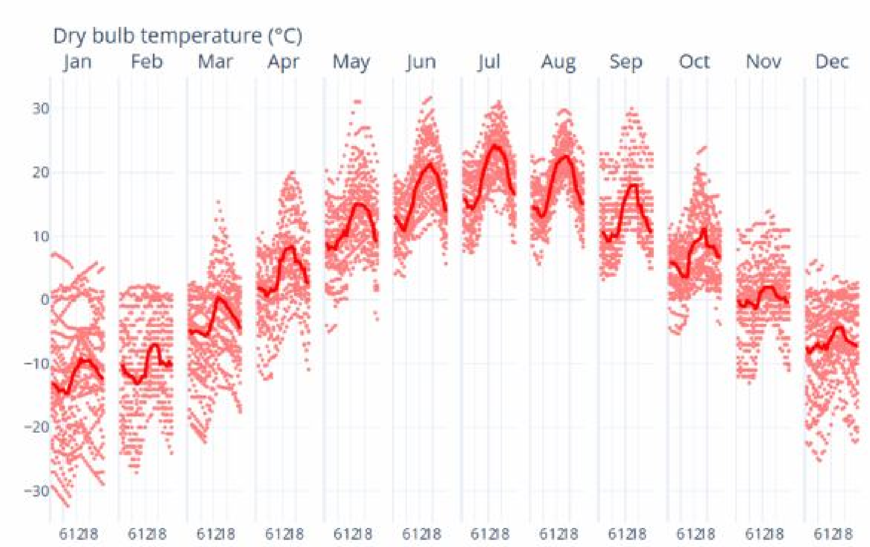
When looking into a different climates for relocation, thermal comfort needs to be considered through dry bulb and humidity.

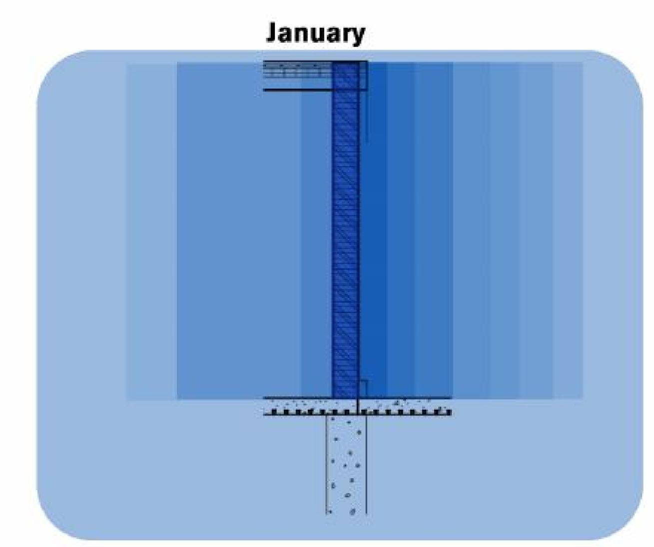
Laorca, Lecco, Italy

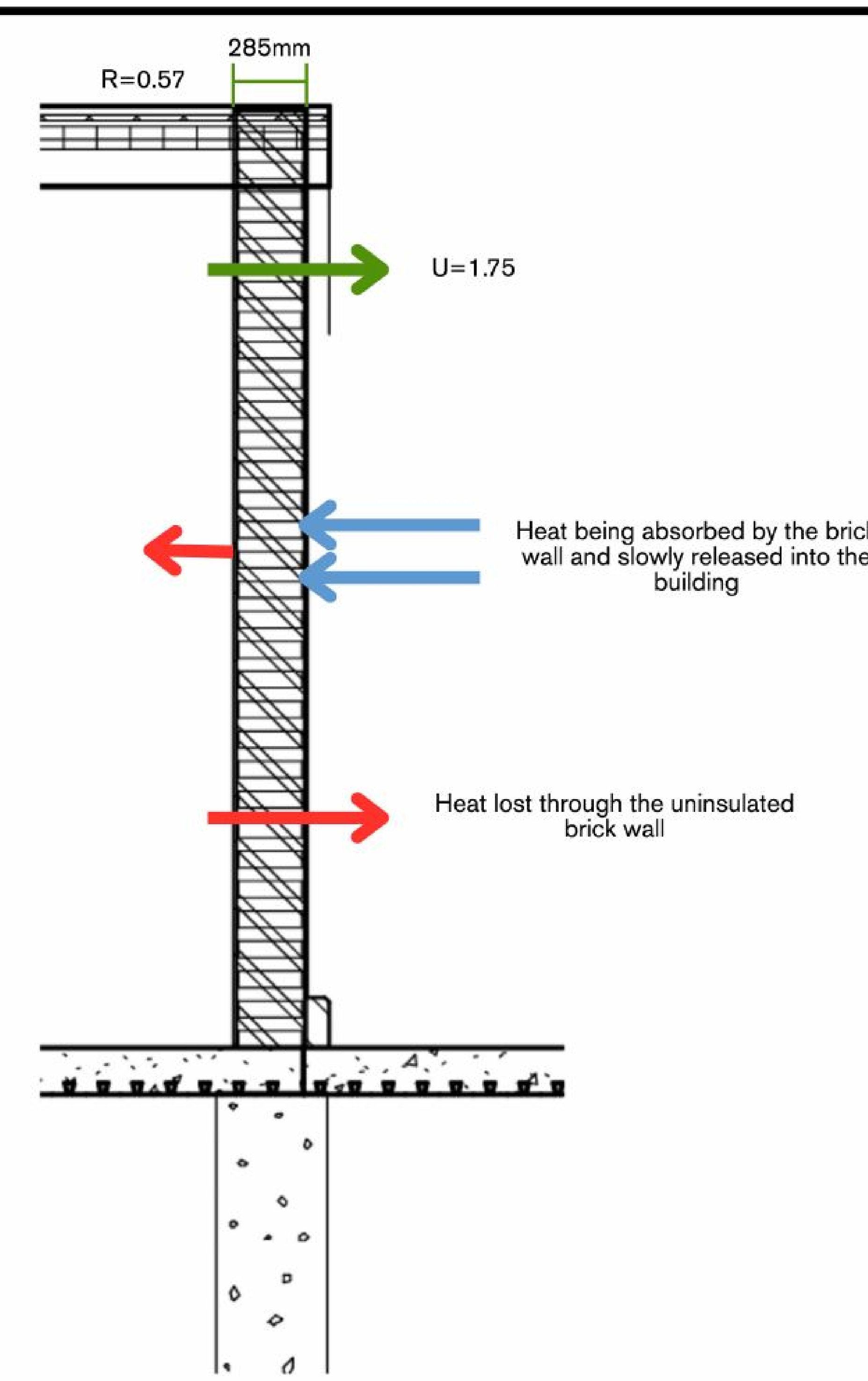
Muuratsalo sits on a forested island in Lake Päijänne, where moisture from the lake and frequent precipitation add to the weathering stresses on brick and masonry. Jyväskylä's Dfc climate also has no dry season, keeping relative humidity high throughout the year. Similarly, Grigna Settentrione, classified as Cfb, experiences frequent precipitation, cloudy skies, and alpine snow. Both settings therefore expose experimental masonry to moisture challenges, whether through Finland's freeze–thaw cycles or Italy's alpine precipitation. In both cases, architects must design with vapor management, drainage, and durable detailing to prevent material deterioration.

Maniwaki, Quebec, Canada

This location was chosen for its similar climate and terrain to the Experimental House's current location. However, this means that Maniwaki, although has warmer summers and a longer warm season then Jyvaskyla, it also has the same freeze-thaw cycle and colder winters. Quebec also experiences excessive snow which the experimental house is not designed to handle.



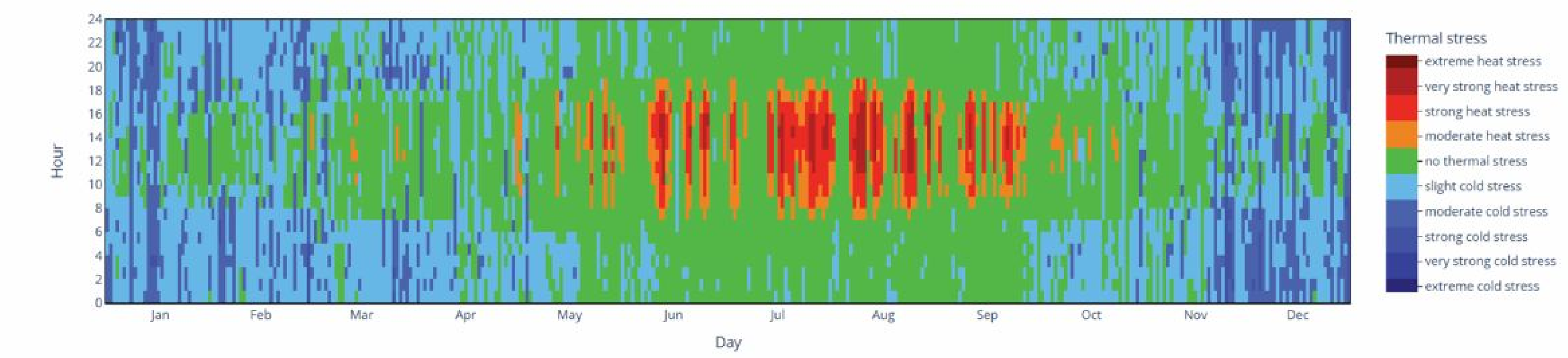


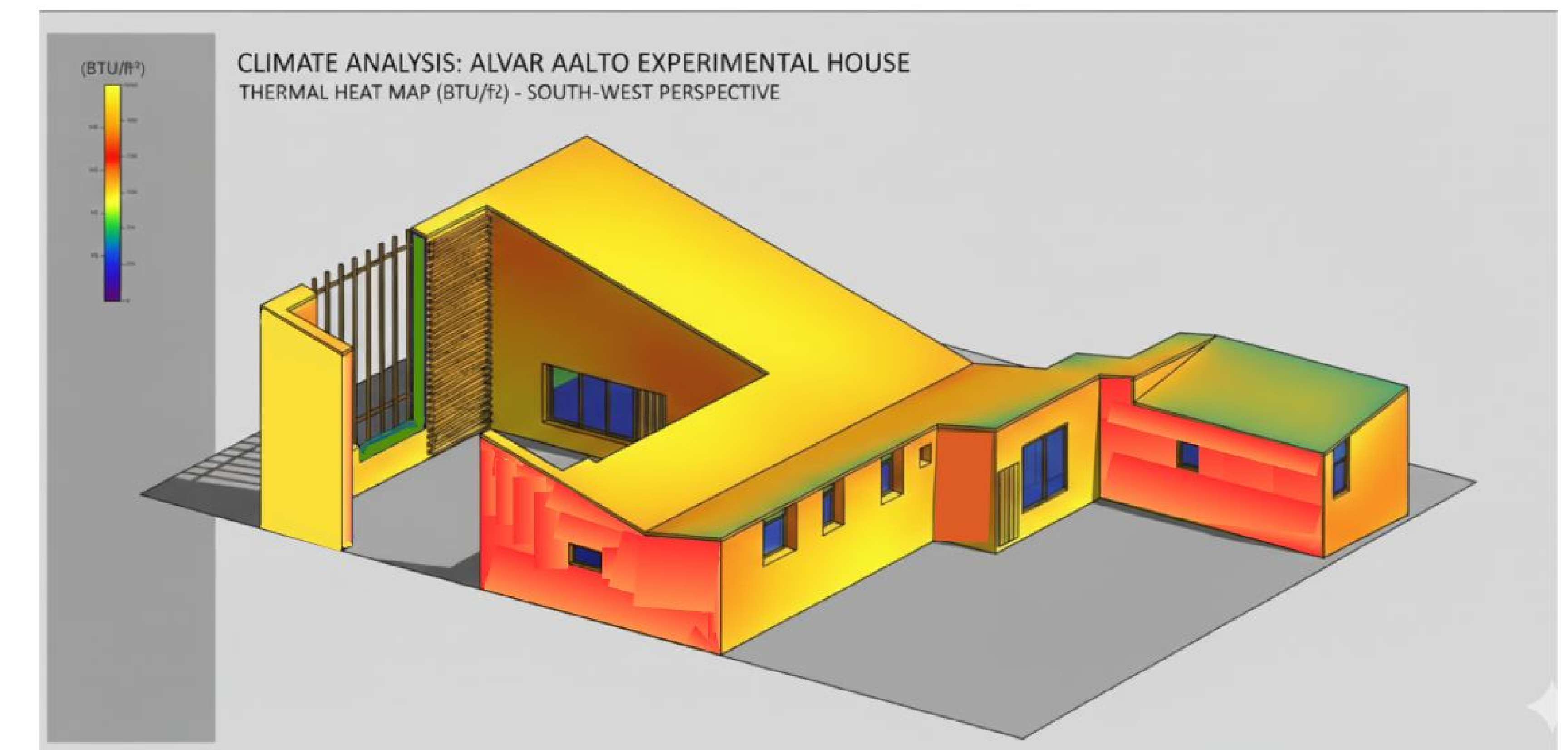


Final Site

Outside Portland, Oregon, USA

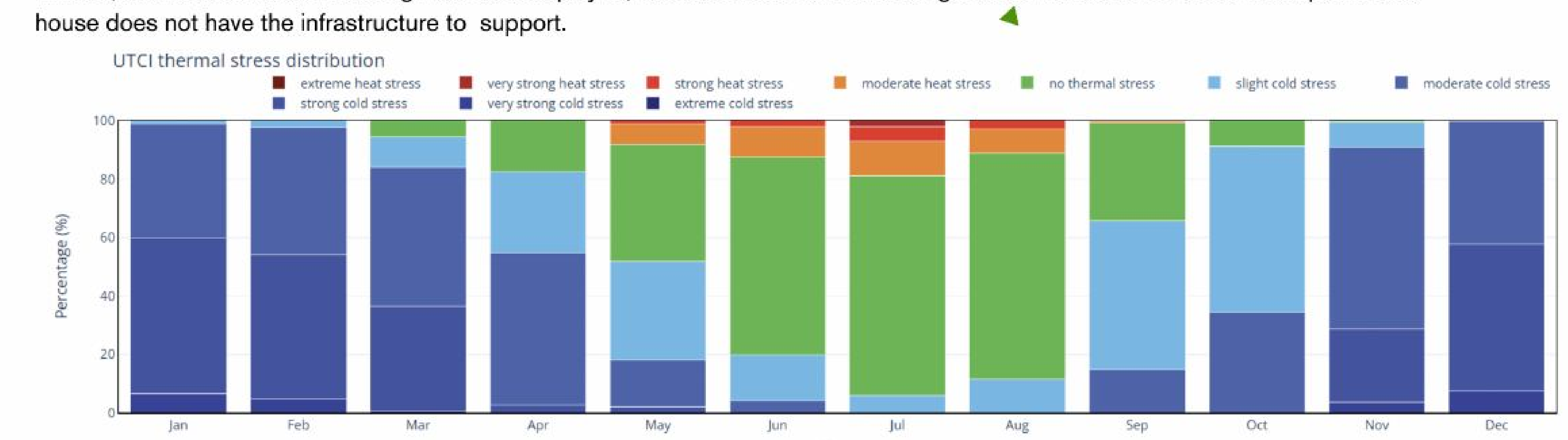
Portland was considered due to its similar geological features to its current location. Focusing on thermal stress and comfort in the experimental house, Portland has a warmer climate making the house usable year round. The lack of insolation will not be a problem in the new location.



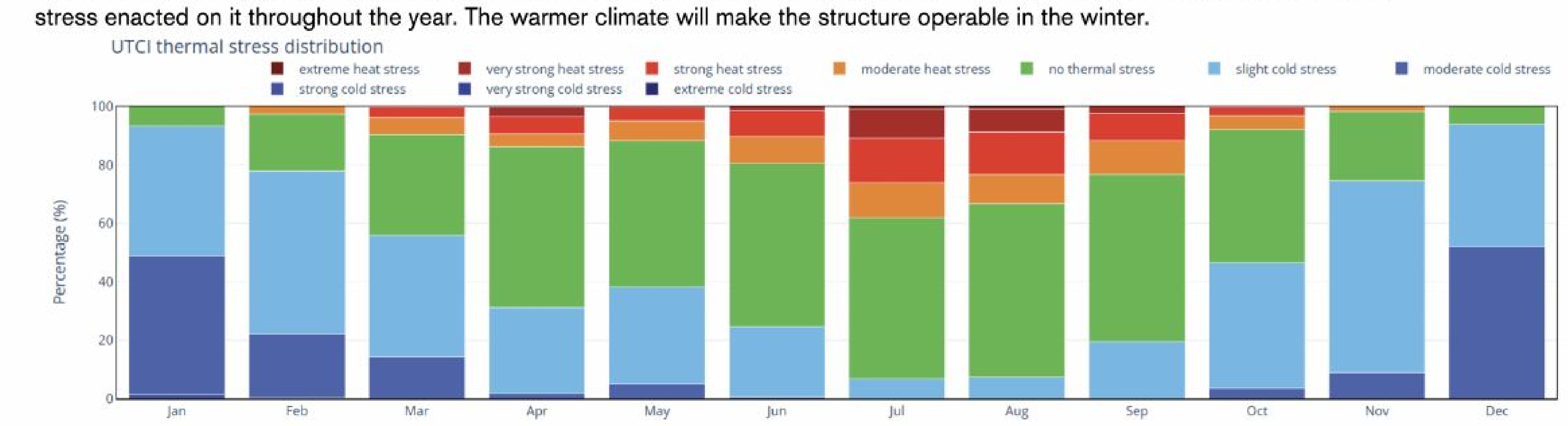


Side-By-Side Comparison of Current and Chosen Location

The charts below showcases the thermal stress the climate will have on a structure in each month. The first chart displays Jyvaskyla, Finland, the current location. Through the data displayed, the current location has a large colder stress load. A load the experimental house does not have the infrastructure to support.



The chart below displays the data from just outside Portland, Oregon. It is clear that the experimental house will have less thermal stress enacted on it throughout the year. The warmer climate will make the structure operable in the winter.



Carleton University Azrieli School of Architecture and Urbanism

EXPERIMENTAL HOUSE
MELALAMMETIE,40900, FINLAND

DATE OF CONSTRUCTION
1952-21954

CLIMATE AND ENVIRONMENTAL RETROFIT

GROUP 7

SHEET NUMBER:
A-701
THERMAL ANALYSIS

TEAM MEMBER:
MANREET ATWAL

REVISION SCHEDULE:

NOORAIZ NOORANI	10/01/2025	CREATED 3D GRAPHICS DISPLAYING THERMAL ANALYSIS
MANREET ATWAL	10/05/2025	RAN WEATHER ANALYSIS FOR EACH CONSIDERED LOCATION USING CLIMA TOOL
MANREET ATWAL	10/05/2025	ANNOTATED GRAPHS AND CHARTS
KAITLIN GIBSON	10/06/2025	ADDED CHART
MANREET ATWAL	10/07/2025	SLIGHT REFORMATTING
MANREET ATWAL	10/14/2025	CREATED 2D GRAPHICS AND ANNOTATED WALL SECTION
MANREET ATWAL	10/14/2025	ANNOTATED 3D GRAPHIC FOR NEW LOCATION

Material Geography

Brick

Extraction and Early Stages

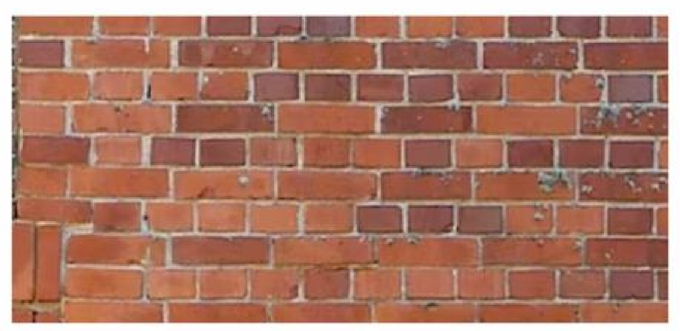
Clay bricks are primarily made up of either one specific type of clay or several mixes. Including surface clays, shales, and fire clays depending on the extraction site.

Storage and Preparation

As the clay is mined, it is continuously being transported to storage areas to ensure the production of bricks even when the pit is not operating. Storing the clay allows for the blending of different mixes to create a more uniform product.

Forming

In order to get the clay into a more workable material after any large, unusable stone pieces have been removed, water is added. This step is called tempering and happens in a mixing chamber. Now the clay is ready to be formed in molds or through a die with an automatic cutter.

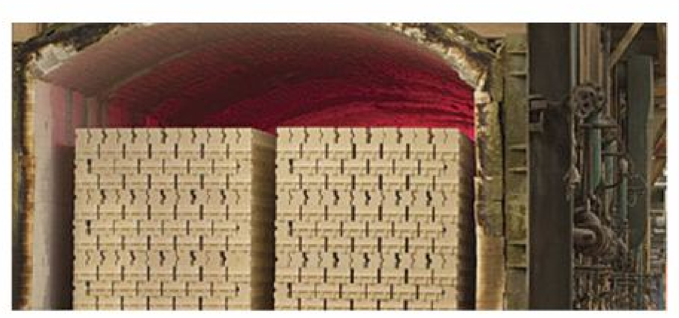


Drying

Depending on the how the brick was formed will determine the final moisture percentage which will determine the length of the drying process. The bricks first have to go through a drying chamber to remove as much moisture as possible before being fired.

Firing

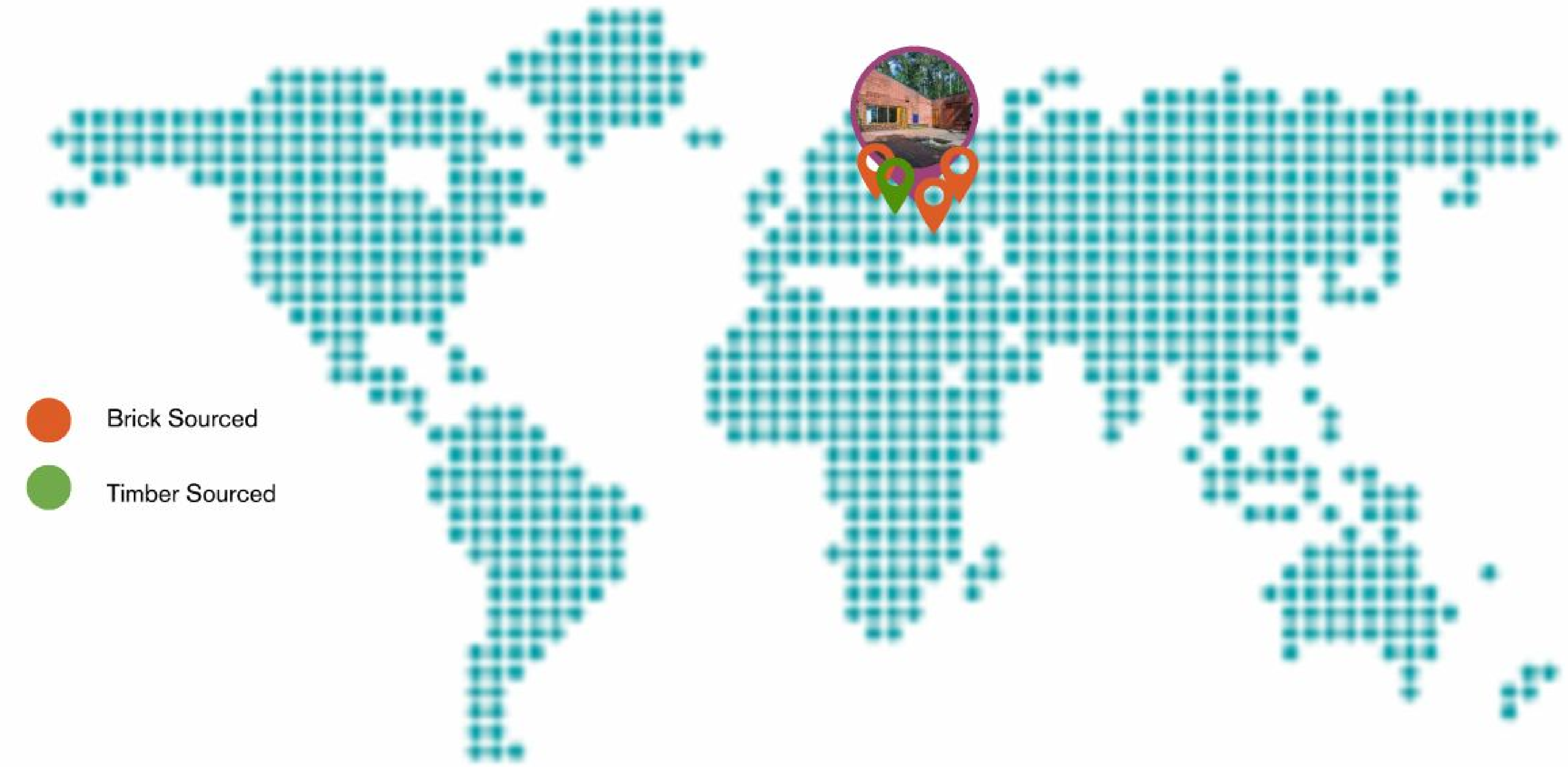
The clay bricks are fired in a kiln from 10 to 40 hours depending on the type of kiln and fuel source. The firing stage is broken down into 5 general stages:
1. final drying; 2. dehydration; 3. oxidation; 4. vitrification; and 5. flashing. The temperature of the kiln continues to rise from stage 1 to 4.

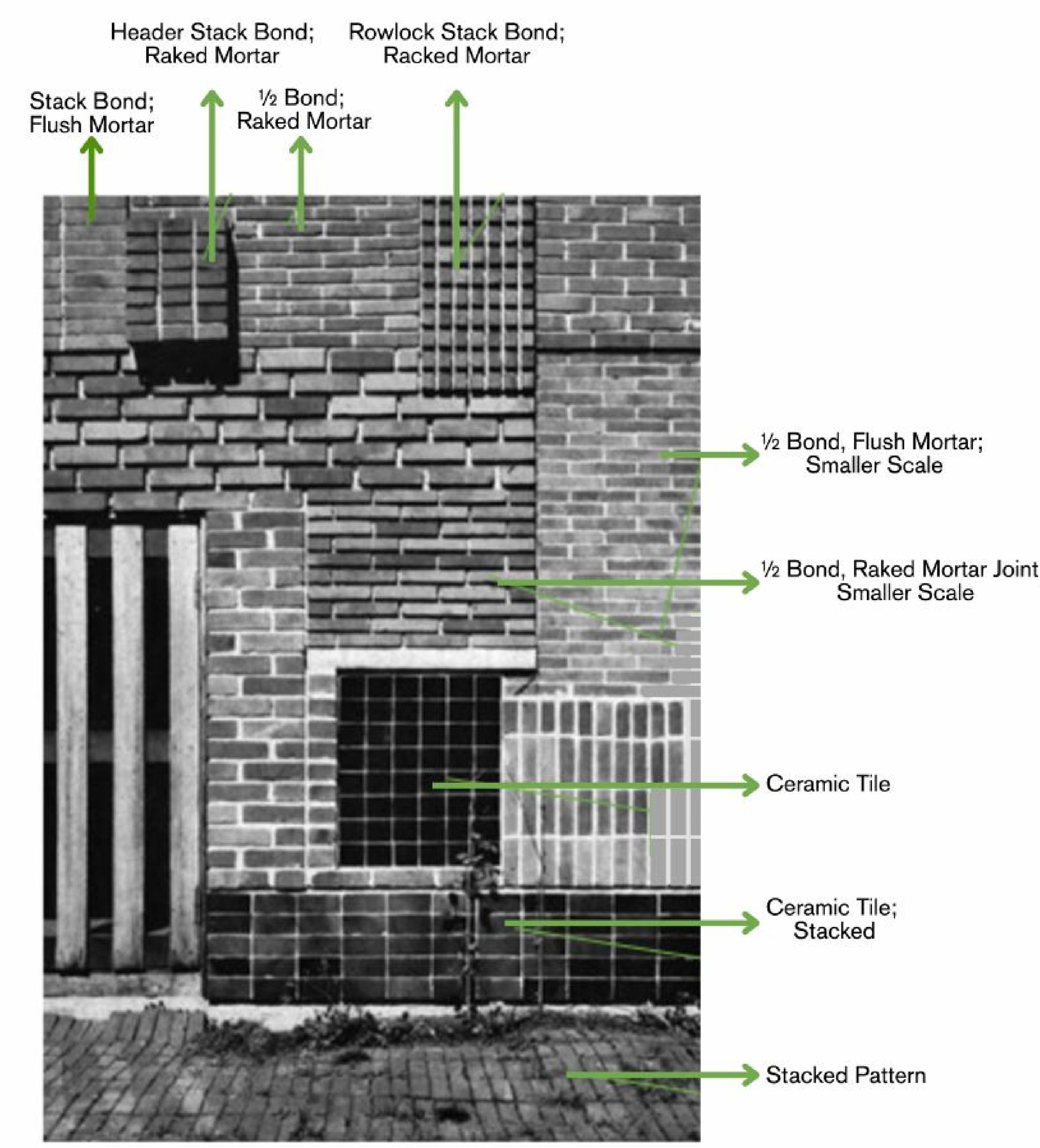


Cooling and Dehacking

Once the bricks are done in the kiln, the cooling process begins. The cooling time can be anywhere from 10 to 24 hours depending on the kiln type. The duration will directly impact the final colour.

The exterior envelope of the home is composed of recycled brick from near by projects in Finland. There is 50 different brick types used through out to the home to test how they would with stand the climate





Timber

Pre-Extraction

Timber is a renewable resource originating from sustainably managed forests, taking decades to grow. Responsible tree growth ensures replanting and ecological biodiversity protection for long term timber supply.

Harvesting

Once grown to their planned size, the trees are cut down using modern mechanical equipment or, on a smaller scale, with chainsaws or traditional hand tools. Depending on the scale, forest type, and location one or more of the above method(s) are used.

Transportation

Once felled, these logs are efficiently transported to sawmills. Exact logistics depend on the situation, although in general, purpose built log transport trucks and boats (if near waterways) are used.

Sawing

After arriving at the sawmill, logs are sawn into different sized beams, planks, or other shapes depending on need and log size. Different sawing methods are used determined by desired dimensions, application, and surface finish.

Drying

The sawn timber are then air dried outdoors or kiln dried inside in a controlled environment. This is to attempt to prevent or minimize shrinking, warping, and other physical deformations.

Grading and Storage

Once dried the timber is mechanically or visually graded based on strength, moisture content, and appearance. It is then organized by grade and stored in controlled, dry conditions.

Final Preparation and Construction

Prior to installation, final modifications to the timber take place, including final cuts, sanding, and planning. Surface treatments are also applied to protect against weather and other environmental factors. Once transported on-site the timber is finally assembled and constructed.

Carleton University Azrieli School of Architecture and Urbanism	EXPERIMENTAL HOUSE MELALAMMETIE,40900, FINLAND DATE OF CONSTRUCTION 1952-21954	CLIMATE AND ENVIRONMENTAL RETROFIT GROUP 7	SHEET NUMBER: A-801 MATERIALS GEOLOGY	TEAM MEMBER: KAITLIN GIBSON AND COLIN MURRAY

REVISION SCHEDULE:

KAITLIN GIBSON	10/04/2025	FORMATED GENERAL LAYOUT
KAITLIN GIBSON	10/05/2025	CONDUCTED RESEARCH IN BRICK ECOLOGY AND COMPLIED INTO CANVA
MANREET ATWAL	10/05/2025	ADDED ANNOTATED IMAGE AND MAP OF SOURCED MATERIALS
KAITLIN GIBSON	10/12/2025	ADDED IMAGES AND UPDATED TEXT AFTER PIN-UP
COLIN MURRAY	10/08/2025 - 10/14/2025	CONDUCTED RESEARCH ON TIMBER ECOLOGY AND COMPLIED INTO CANVA
COLIN MURRAY	10/14/2025	ADDED TIMBER IMAGES AND TEXT TITLES